

Topology Optimization of Heat Transfer Applications Operating in the Turbulent Regime

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Preface

This thesis marks the culmination of my Master of Science in Mechanical Engineering at KU Leuven. The journey of exploring computational fluid dynamics and mathematical optimization has been both intensely challenging and deeply rewarding. Bridging the gap between theoretical mathematics and practical engineering applications required countless hours of coding, debugging, and analyzing flow fields, but seeing the optimizer successfully carve out organic, aerodynamic fluid channels made the entire process worthwhile.

I would like to express my sincere gratitude to my supervisors, Prof. Niels Horsten and Prof. Maarten Blommaert, for providing me with the opportunity to work on this fascinating topic. Their academic guidance, critical insights, and continuous support were instrumental in shaping the direction and rigor of this research. A very special thanks goes to my assistant-supervisor, Esteban Foronda Obando, for his day-to-day patience, technical expertise, and willingness to help me navigate the complex numerical hurdles within FEniCS.

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Tiebert Lefebure

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Abstract

As modern electronic and automotive systems grow increasingly compact and powerful, efficient thermal management has emerged as a fundamental engineering challenge. While topology optimization has proven to be a highly effective computational design tool for the fluid manifolds that drive these cooling systems, traditional methodologies rely heavily on laminar flow approximations. These laminar assumptions severely limit industrial applicability, as they fail to capture the separation, curvature effects, and turbulence-model dependence inherent to high-speed internal flows.

This thesis develops and implements a density-based topology optimization framework for turbulent internal flows as a baseline for future heat-transfer applications. The computational methodology is built upon the open-source FEniCS platform and utilizes the fictitious domain method with Brinkman momentum penalization to conceptualize solid boundaries within a continuous fluid mesh. To capture the complex physics of high-Reynolds flows, the framework integrates the one-equation Spalart-Allmaras (SA) turbulence model. The optimization process is driven by a frozen-turbulence continuous adjoint, meaning that the eddy-viscosity and wall-distance fields are updated in the forward loop but treated as fixed coefficients during sensitivity assembly.

The assessment strategy is deliberately separated into three stages. First, the predictive scope of the SA model for curvature-dominated internal flows is established and benchmarked against Ansys Fluent. Second, the custom SA implementation in FEniCS is verified on turbulent benchmark flows. Third, the Alexandersen pipe-bend optimization is verified through finite-difference sensitivity checks and exported to Ansys Fluent for sharp-geometry validation. This final stage compares FEniCS-SA with Ansys-SA on the same geometry and then evaluates the same design with SST $k-\omega$ to quantify turbulence-model sensitivity. In this revised framing, laminar-versus-turbulent design comparison is retained as supporting context rather than treated as CFD validation in itself.

The resulting thesis is therefore intentionally conservative in its claims. Its central question is not merely whether turbulent topologies look different from laminar ones, but whether SA-driven topology optimization produces low-loss designs whose predicted performance remains credible under independent CFD re-analysis. This establishes a more defensible and transparent baseline for future three-dimensional and conjugate heat-transfer extensions.

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List of Abbreviations and Symbols

Abbreviations

CAD	Computer-Aided Design
CEV	Constant Eddy Viscosity
CFD	Computational Fluid Dynamics
CG1	Continuous Galerkin (Linear Basis Functions)
CHT	Conjugate Heat Transfer
DG0	Discontinuous Galerkin (Piecewise Constant Elements)
DNS	Direct Numerical Simulation
FEM	Finite Element Method
FVM	Finite Volume Method
IPCS	Incremental Pressure Correction Scheme
MMA	Method of Moving Asymptotes
PDE	Partial Differential Equation
RAMP	Rational Approximation of Material Properties
RANS	Reynolds-Averaged Navier-Stokes
SA	Spalart-Allmaras
SIMP	Solid Isotropic Material with Penalization
SST	Shear Stress Transport
SUPG	Streamline Upwind Petrov-Galerkin
TO	Topology Optimization
UFL	Unified Form Language

Symbols

Fluid Dynamics & Turbulence

d, y	Distance to the nearest wall / Hydraulic diameter [m]
De	Dean number [-]
k	Turbulent kinetic energy [m^2/s^2]
$p, \bar{p}, p'$	Instantaneous, mean, and fluctuating static pressure [Pa]
ΔP	Macroscopic total pressure drop [Pa]
R_c	Radius of curvature [m]
Re	Bulk Reynolds number [-]
$S, \tilde{S}$	Scalar vorticity magnitude and modified SA vorticity magnitude [$1/\text{s}$]
t	Time / Pseudo-time [s]
$\mathbf{u}, \bar{u}_i$	Velocity vector / Mean velocity component [m/s]
u'_i	Fluctuating velocity component [m/s]
u_τ	Friction velocity [m/s]
y^+	Non-dimensional wall distance [-]
μ, μ_t	Dynamic and turbulent eddy viscosity [Pa·s]
ν, ν_t	Kinematic and turbulent eddy kinematic viscosity [m^2/s]
$\tilde{\nu}$	Spalart-Allmaras working variable [m^2/s]
ρ	Fluid density [kg/m^3]
Ω_{ij}	Vorticity tensor [$1/\text{s}$]
κ	von Kármán constant [-]

Numerical Implementation (FEniCS)

G, G_0	Relaxed Eikonal field variable and boundary value [$1/\text{m}$]
h, h_{avg}	Finite element cell diameter and average cell size [m]
$\ \cdot\ _{L_2}$	L_2 (Euclidean) norm [-]
$[[\cdot]]$	Jump operator across an internal element facet [-]
$\mathbf{n}$	Outward-pointing normal vector [-]
v, ξ	Test functions for pressure and turbulence equations [-]
$\alpha_u, \alpha_{\tilde{\nu}}$	Relaxation factors for velocity and turbulence [-]
Δt	Pseudo-time step [s]
ϵ_{tol}	Convergence tolerance [-]
ϕ	General scalar field variable [-]
σ_w	Eikonal relaxation constant [-]
τ	SUPG stabilization time parameter [s]

Topology Optimization

$\alpha(\gamma)$	Continuous Brinkman penalization term / inverse permeability
$\alpha_{solid}, \alpha_{fluid}$	Maximum solid and minimum fluid Brinkman penalties
α_{dg}	Artificial diffusion scaling factor for DG0 filter [-]
β	Heaviside projection steepness parameter [-]
γ, γ_{raw}	Continuous and raw design variable (1 for fluid, 0 for solid) [-]
$\tilde{\gamma}, \bar{\gamma}$	Filtered and Heaviside-projected design variables [-]
η	Heaviside projection threshold [-]
J	Objective function (Power Dissipation) [W]
q	RAMP penalization tuning parameter [-]
r	Continuous Helmholtz filter radius [m]
$\mathbf{s}, \mathbf{s}^*$	Monolithic forward state and continuous adjoint state vectors
V_f, V_{domain}	Physical fluid volume and total design domain volume [m ³]
V_{max}	Maximum allowable fluid volume fraction constraint [-]
$\Omega, \partial\Omega, \Gamma_{int}$	Computational domain, domain boundary, and internal element facets

Chapter 1

Introduction

1.1 Background and Motivation

With the continued miniaturization of components in the electronics and automotive industries, thermal management has become a critical bottleneck in the development of new technologies. While the heat sinks themselves are vital, the efficiency of any cooling system relies heavily on the fluid manifolds that deliver the coolant. If the flow distribution is uneven or if pumping the fluid requires excessive power due to massive pressure drops, the overall efficiency of the system severely degrades. Addressing this fluid delivery problem from an engineering perspective is a challenging task due to the complex aerodynamic behaviors that govern high-speed flow networks.

Innovative design strategies based on mathematical optimization have made it possible to improve the performance of these devices while capturing the complexity of the underlying physical phenomena. Among these strategies, topology optimization has emerged as a highly effective computational design tool for advanced thermal management [19]. For instance, the geometry of a complex microfluidic channel network can be automatically optimized to distribute coolant with specified flow rates without relying on human intuition [51]. Furthermore, the ultimate frontier of this technology lies in the design of conjugate and two-fluid heat exchangers, where the algorithm must simultaneously maximize heat transfer while strictly limiting the fluid pumping power [26]. However, before multi-physics heat transfer can be reliably optimized at an industrial scale, the fundamental algorithms governing the turbulent fluid dynamics must first be stabilized and validated.

1.2 Conceptual Overview of Density-Based Topology Optimization

To understand the challenges of applying optimization to turbulent fluids, it is first necessary to distinguish topology optimization from traditional design methodologies. Historically, computational design relied on size optimization (altering the thickness of predefined components) or shape optimization (moving the boundary

nodes of a predefined geometry). While effective, these methods are inherently limited by their starting topology. A shape optimization algorithm cannot spontaneously create a new hole or split a single fluid channel into two.

Density-based topology optimization, pioneered by Bendsøe and Kikuchi in 1988 for solid mechanics [12], approaches the problem fundamentally differently. Instead of moving boundaries, the design space (the computational domain) is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the specific convention adopted for this thesis, a value of $\gamma = 1$ represents a pure fluid channel (e.g., the cooling air or water), while $\gamma = 0$ represents a solid, impermeable structure (e.g., a manifold wall).

However, because gradient-based optimizers require continuous and differentiable functions, this discrete binary choice is mathematically relaxed. The design variable is allowed to take intermediate "gray" values between 0 and 1. To give physical meaning to these intermediate states, the fluid equations are modified using a fictitious porous medium approach, where intermediate densities correspond to varying levels of flow permeability. The optimization algorithm iteratively updates the γ field (as conceptualized in Figure 1.1) to minimize a specific objective function (e.g., dissipated power) subject to physical constraints (e.g., a maximum allowable fluid volume fraction). The choice of the initial value for γ is a critical parameter for solver stability. A uniform initial guess of $\gamma = 0.5$ is standard practice to ensure an unbiased algorithmic search across the design space.

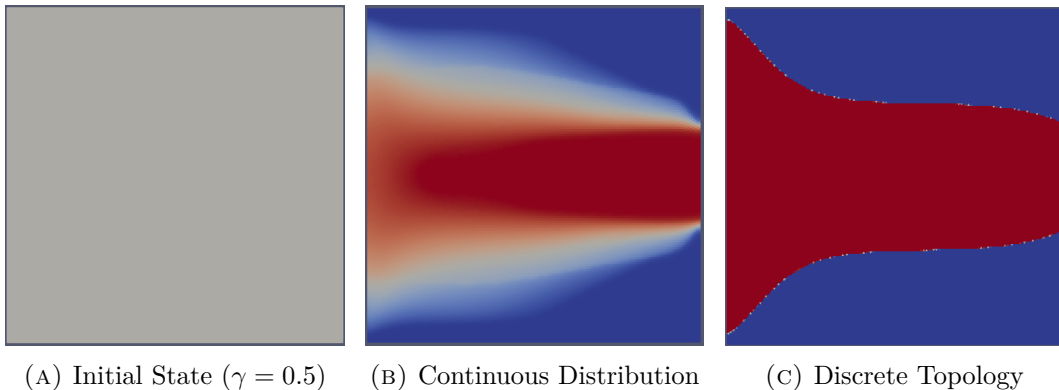


FIGURE 1.1: Iterative density-based topology optimization process: (a) initialization with a uniform field, (b) intermediate continuous distribution where $\gamma \in [0, 1]$, and (c) final discrete fluid topology where $\gamma \in \{0, 1\}$.

1.3 The Evolution of Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent.

The optimization of fluid manifolds remains an active and highly challenging area of research [19]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [13]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [21]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, successfully adapting the methodology to standard incompressible Navier-Stokes equations [16, 3, 51].

However, the assumption of laminar flow significantly simplifies the governing equations and the subsequent derivation of the adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to most industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the fully turbulent regime [48, 17, 49].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with complex Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear behaviors, and severe numerical instabilities during the iterative optimization process. To mitigate this, gradient-based turbulent optimizations frequently rely on the "frozen turbulence" assumption, where the turbulent viscosity field is held constant during the sensitivity analysis to avoid differentiating the complex turbulence transport equations. While this assumption provides necessary solver stability and has driven massive success in industrial shape optimization for automotive aerodynamics [31], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing and highly complex mathematical challenge.

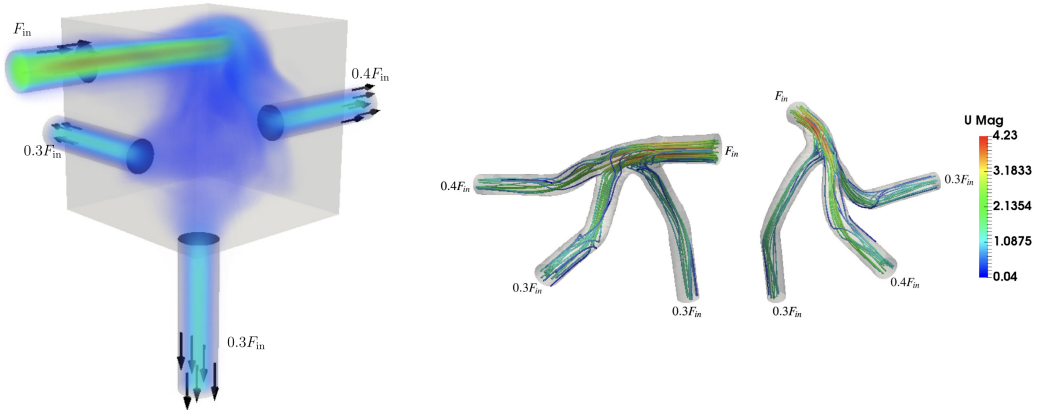


FIGURE 1.2: Topology optimized manifolds with turbulent flow. Adapted from Dilgen et al. [17].

Pioneering works by Yoon (2016) [48] and Dilgen et al. (2018) [17] have pushed

the boundaries of these turbulent frameworks. As seen in Figure 1.2, integrating advanced turbulence models allows the optimizer to generate highly complex, organic fluid manifolds specifically tailored to manipulate and guide turbulent flow distributions efficiently.

1.4 Objectives and Research Questions

The main objective of this Master’s thesis is to extend the application of topology optimization to the design of fluid manifolds operating strictly in the turbulent regime. To achieve this goal, this work investigates the coupling of fluid dynamics and turbulence modeling within an automated mathematical optimization framework. By strictly isolating the fluid mechanics from complex thermodynamic calculations, this thesis bounds its scope to establish a highly stable and mathematically rigorous baseline for pure aerodynamic manifold design.

Rather than relying solely on black-box commercial Computational Fluid Dynamics (CFD) software, the core of this thesis involves implementing the framework using **FEniCS**, an open-source computing platform for solving partial differential equations [4]. FEniCS, combined with high-level discrete adjoint methods (e.g., **dolfin-adjoint** [5]), provides the mathematical flexibility required to implement, verify, and integrate the one-equation Spalart-Allmaras (SA) turbulence model into a fluid-design loop. For full transparency and reproducibility, the complete FEniCS source code developed for this thesis is open-source and publicly available at: <https://github.com/TiebertLefebure/TurbulentT0.git>.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of re-analyzing the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with alternative RANS closures.

Based on this methodology, this thesis seeks to answer the following specific research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curvature-dominated internal turbulent flows, and where do its physical limitations appear?**
2. **Can a custom standard-Galerkin FEniCS implementation of the Spalart-Allmaras turbulence model converge robustly and achieve quantitative agreement with established commercial CFD solvers?**
3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis, both with the same SA model and with alternative RANS closures?**

1.5 Outline of the Thesis

The remainder of this thesis is structured into the following five chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, the principles of RANS modeling, and the physical characteristics of the Spalart-Allmaras model.
- **Chapter 3 (Implementation and Verification of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the chosen turbulence model within the FEniCS framework, including fractional-step solvers, standard Galerkin discretization, convergence controls, and quantitative benchmark validations.
- **Chapter 4 (Topology Optimization for Turbulent Flows: Methodology):** Introduces the density-based optimization mathematical framework. It details the Brinkman momentum penalization, the power-law SA and wall-distance penalties, the reciprocal penalized wall-distance equation, and the rationale for adopting the frozen turbulence assumption as the baseline design engine.
- **Chapter 5 (Topology Optimization for Turbulent Flows: Results, Cross-Code Verification, and Model Sensitivity):** Presents the turbulent topology optimization results, documents the export of the final designs to Ansys Fluent, compares FEniCS-SA against Ansys-SA on the same geometries, and studies sensitivity to alternative turbulence models such as SST $k-\omega$ and standard $k-\epsilon$.
- **Chapter 6 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines the remaining limitations, with particular emphasis on turbulence-model dependence, geometry extraction, and future multi-physics heat-transfer applications.

Chapter 2

Turbulence Modeling

2.1 Introduction to Fluid Dynamics and Turbulence

Fluid dynamics is governed by the Navier-Stokes equations, which describe the conservation of mass and momentum for a viscous continuum fluid [46]. In this introductory text, these equations are presented using index notation (Einstein summation convention) to clearly illustrate the individual tensor components. In this convention, the indices i and j represent the three spatial dimensions (1, 2, and 3, corresponding to the x , y , and z axes). Consequently, x_i denotes the Cartesian coordinates, and u_i represents the corresponding components of the velocity vector. A repeated index within a single term implies a mathematical summation over all three spatial directions. (Note: In subsequent chapters dealing with finite element implementation, this notation will shift to standard vector calculus to align with FEniCS unified form language syntax). The fundamental principle of mass conservation is expressed by the continuity equation. For a general, compressible fluid, this is defined as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (2.1)$$

where ρ is the fluid density. However, for most liquid cooling applications and low-speed fluid manifolds, the fluid can be safely assumed to be incompressible. Under the incompressible flow assumption, the density ρ is constant in both space and time. Therefore, the time derivative of density becomes zero, and the constant density can be factored out of the spatial derivative. Dividing the equation by ρ yields the simplified, incompressible continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (2.2)$$

Similarly, for an incompressible, Newtonian fluid with a constant dynamic viscosity μ , the conservation of momentum is expressed as:

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \quad (2.3)$$

where p represents the static pressure of the fluid. In principle, solving Equation 2.3 directly yields the exact flow field. This approach is known as Direct Numerical Simulation (DNS). However, as the Reynolds number (Re) increases, the flow transitions into a chaotic turbulent state characterized by a broad spectrum of spatial and temporal scales [33]. Resolving the smallest turbulent eddies requires extremely fine spatial grids. For industrial applications, including the design of electronics cooling manifolds, DNS is computationally prohibitive. Therefore, engineers rely on turbulence modeling to predict the averaged effects of these fluctuations without resolving the complete turbulent spectrum [44, 6].

2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The most common approach to turbulence modeling in engineering is the Reynolds-Averaged Navier-Stokes (RANS) framework, based on the theories originally introduced by Osborne Reynolds [33]. The core principle of RANS is the Reynolds decomposition. This mathematical operation separates the instantaneous flow variables into a time-averaged mean component ($\bar{u}_i, \bar{p}$) and a fluctuating component (u'_i, p'):

$$u_i = \bar{u}_i + u'_i \quad \text{and} \quad p = \bar{p} + p' \quad (2.4)$$

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho \left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right) \quad (2.5)$$

The averaging process introduces a new term, $-\rho \overline{u'_i u'_j}$, known as the Reynolds stress tensor. This tensor represents the momentum transfer caused by turbulent fluctuations. The introduction of this term creates a fundamental mathematical closure problem because the RANS equations now contain more unknowns than available physical equations [44].

2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

To close the system and solve for the mean flow, most RANS models rely on the Boussinesq eddy viscosity hypothesis [33]. This hypothesis assumes that the turbulent momentum transfer is analogous to molecular momentum transfer. It relates the Reynolds stresses linearly to the mean velocity gradients using a proportionality constant called the turbulent eddy viscosity (μ_t):

$$-\rho \overline{u'_i u'_j} = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} \quad (2.6)$$

where k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. The primary task of any Boussinesq-based turbulence model is to calculate this local eddy viscosity. It is crucial to highlight a fundamental physical limitation of Equation 2.6: the Boussinesq hypothesis inherently assumes that μ_t is an *isotropic* scalar quantity. It mathematically enforces that turbulence behaves identically in all spatial directions and aligns perfectly with the mean strain rate. While this assumption holds for simple shear flows, it systematically fails in predicting anisotropic phenomena, such as the strong secondary flows induced by streamline curvature.

2.3 Boundary Layer Theory and Near-Wall Treatment

Before selecting a specific turbulence equation to calculate μ_t , the physical structure of a turbulent boundary layer must be considered, as the model must accurately resolve these near-wall regions. The boundary layer is traditionally divided into three distinct regions based on the non-dimensional wall distance (y^+) and non-dimensional velocity (u^+):

$$y^+ = \frac{yu_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \quad (2.7)$$

where y is the absolute distance from the wall, ν is the kinematic viscosity (μ/ρ), and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity derived from the wall shear stress (τ_w) [33].

1. **The viscous sublayer** ($y^+ < 5$): Immediately adjacent to the wall, turbulent fluctuations are damped out by kinematic constraints. Viscous shear dominates, and the velocity profile is strictly linear ($u^+ = y^+$).
2. **The buffer layer** ($5 < y^+ < 30$): A transition region where both viscous and turbulent shear stresses are of equal magnitude.
3. **The log-law region** ($y^+ > 30$): Farther from the wall, turbulence dominates, and the velocity follows a logarithmic distribution predicted by the von Kármán constant [44].

High-Reynolds models (like the standard $k - \epsilon$ model) use empirical "wall functions" to bridge the gap between the wall and the log-law region, requiring the first computational mesh element to be placed at $y^+ > 30$. However, for topology optimization, where solid-fluid boundaries continuously evolve during the optimization loop, empirical wall functions introduce severe mathematical discontinuities and numerical instabilities. Therefore, a "wall-resolved" model that integrates the governing equations all the way through the viscous sublayer ($y^+ \approx 1$) is strictly required for robust gradient-based optimization.

2.4 The Spalart-Allmaras One-Equation Model

While two-equation models (such as $k - \epsilon$ or $k - \omega$) solve separate transport equations for the turbulent kinetic energy and a turbulent length scale, they introduce highly

non-linear source terms. These terms frequently cause numerical instabilities during the adjoint sensitivity analysis required for topology optimization. To achieve a trade-off between physical accuracy and computational efficiency, this thesis utilizes the Spalart-Allmaras (SA) model, originally developed in 1992 for aerodynamic flows with adverse pressure gradients [36]. The SA model is a one-equation model that solves a single transport equation directly for a working variable, $\tilde{\nu}$, which is subsequently linked to the true kinematic eddy viscosity ($\nu_t = \mu_t/\rho$) [37]. The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \quad (2.8)$$

The terms on the right-hand side represent the production, destruction, and diffusion of the turbulent working variable, respectively. Crucially, the production term relies on a modified vorticity scalar, $\tilde{S}$. This indicates that turbulence generation in the SA model is fundamentally driven by fluid shear, a physical characteristic that introduces unique boundary-condition sensitivities during advanced adjoint calculations (discussed further in Chapter 4). Furthermore, the destruction term is explicitly a function of d , the absolute distance to the nearest wall. This makes the SA model a highly effective "wall-resolved" low-Reynolds model. It is inherently sensitive to near-wall damping effects and integrates smoothly through the viscous sublayer without requiring complex empirical wall functions or a second length-scale equation [34]. Detailed derivations of the model constants and wall damping functions are deferred to Chapter 3, where the numerical implementation in FEniCS is discussed.

2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. As it lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption, it struggles to capture the anisotropic physics of streamline curvature. When fluid passes through a bend, centrifugal forces induce secondary flows perpendicular to the main flow direction. To quantify the strength of these secondary flows, the dimensionless Dean number (De) is utilized:

$$De = Re_d \sqrt{\frac{d}{2R_c}} \quad (2.9)$$

where d is the characteristic length of the cross-section (e.g., the pipe diameter D or hydraulic diameter D_h), R_c is the radius of curvature, and Re_d is the bulk Reynolds number based strictly on that exact same characteristic length. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

A review of the literature demonstrates that the predictive capability of the SA model is heavily dependent on this Dean number. For mildly curved flows, the SA model performs relatively well. Apalowo [8] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9,000$ (operating at $Re_D \approx 34,000$, based on the pipe diameter). Under these moderate conditions, where the bend radius was seven times larger than the pipe diameter ($R_c = 7D$), the SA model provided a highly acceptable approximation of the streamwise velocity profiles compared to higher-order models, successfully capturing the primary flow physics.

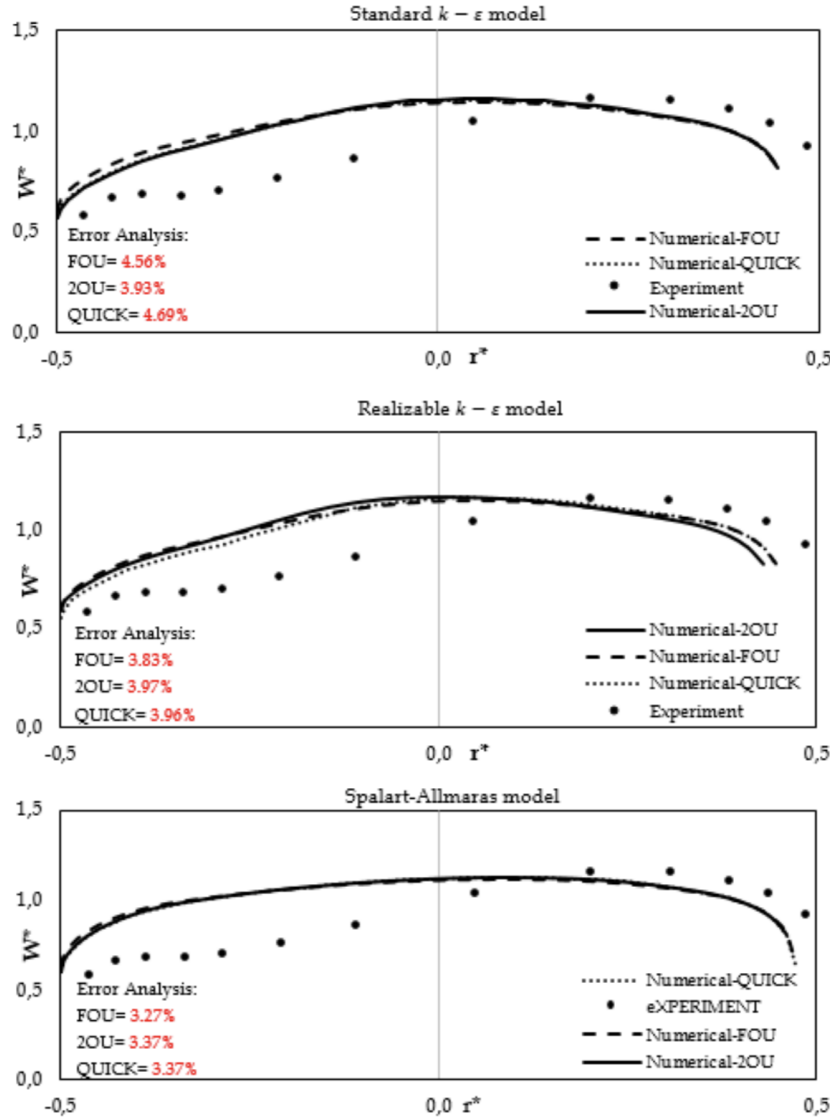


FIGURE 2.1: Streamwise velocity profiles in a 90° pipe bend ($R_c = 7D$, $Re_D \approx 34,000$, $De \approx 9,000$). The SA model performs relatively well, closely matching the experimental data and two-equation models for mild curvature. Reprinted from Apalowo [8].

However, as the curvature tightens and the Dean number increases, the limitations of the isotropic Boussinesq assumption become severe. Abuhatira et al. [1] evaluated the internal turbulent flow through a 90° pipe bend characterized by a tighter curvature ratio of $R_c = 2.0D$. Operating at a bulk Reynolds number of $Re_D = 60,000$, this geometry generated strong centrifugal forces and a corresponding Dean number of $De \approx 30,000$. Unlike extreme benchmark cases where all isotropic models fail uniformly, this configuration distinctly isolates the shortcomings of the Spalart-Allmaras formulation. At the critical bend exit ($X/D = 0$), the SA model drastically mischaracterized the near-wall velocity profiles, completely missing the flow physics close to the inner wall that higher-order two-equation models (such as $k-\epsilon$ and $k-\omega$) resolved with much greater accuracy. This confirms that the failure is not attributed to inertial scaling, but is strictly driven by the model's inability to compensate for strong anisotropic centrifugal effects.

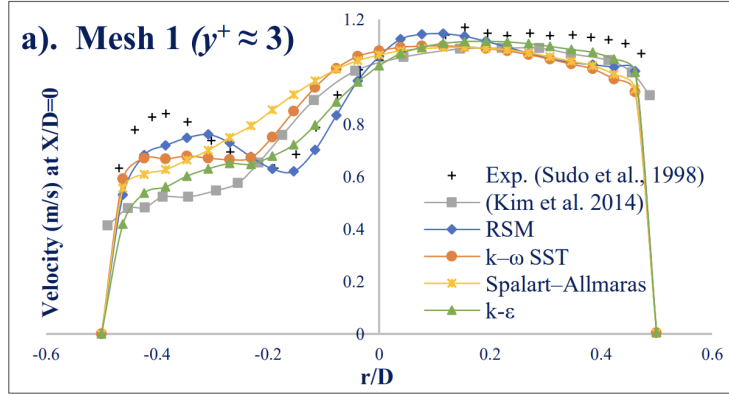


FIGURE 2.2: Turbulence model predictions at the 90° bend exit ($X/D = 0$, $R_c = 2.0D$, $Re_D = 60,000$, $De \approx 30,000$). Due to the strong curvature, the standard Spalart-Allmaras model fails to accurately capture the near-wall velocity profile compared to two-equation closures. Adapted from Abuhatira et al. [1].

While the literature clearly establishes that the SA model is viable at $De \approx 9,000$ and fails to capture inner-wall physics at $De \approx 30,000$, a bracketed or case-specific threshold between these states must be estimated to justify its use in topology optimization. Therefore, to better evaluate where the model begins to break down, rigorous benchmark studies were subsequently conducted within Ansys Fluent.

2.6 Benchmark Studies in Ansys Fluent

To investigate where the SA model's accuracy deteriorates, benchmark simulations were performed using **Ansys Fluent**. The study focused on two distinct, fully three-dimensional (3D) internal flow geometries: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized and rigorously tested numerical baseline, the computational meshes, fluid properties, and boundary conditions for these simulations were adopted directly from the official Ansys Fluid Dynamics Verification

Manual [7]. Specifically, the 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. These cases evaluate industrial flow regimes ($Re_D > 40,000$) to ensure the turbulence model is rigorously verified against dominant inertial physics before being implemented into the custom FEn-iCS solver. To evaluate the accuracy of the one-equation model, the results were compared directly against the $k - \omega$ SST model, an industry standard for accurately predicting adverse pressure gradients and flow separation.

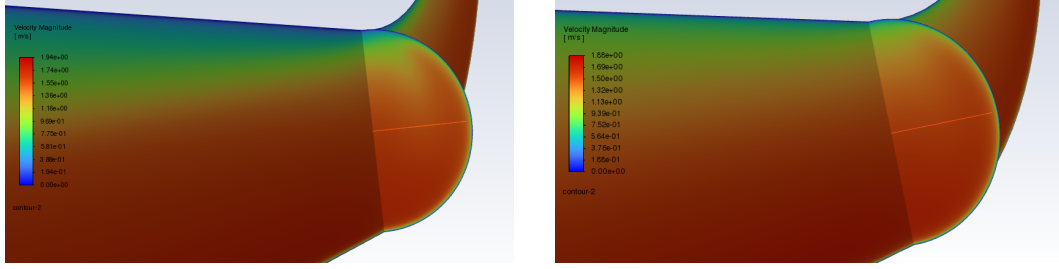
2.6.1 Mild Curvature: The U-Bend Case

The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$. Consistent with the findings of Apalowo for moderate bends, the SA model performed well under these conditions. As shown in Figure 2.3, the qualitative velocity contours of the SA model accurately replicate the bulk flow distribution of the $k - \omega$ SST model. Furthermore, the quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [41].

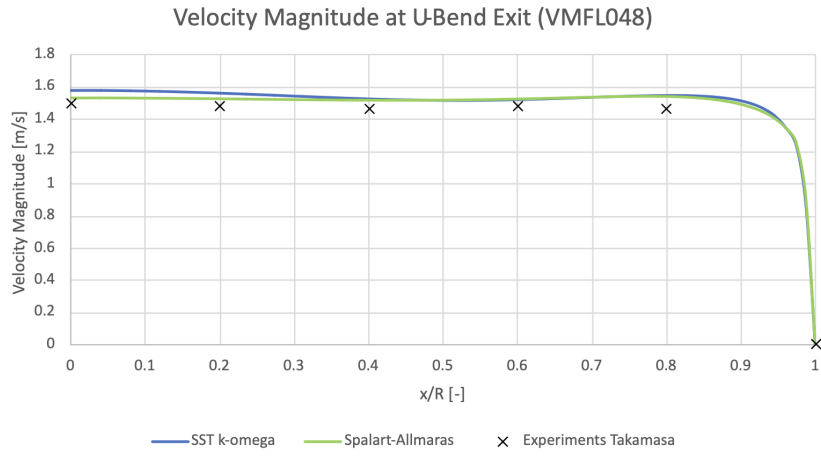
2.6.2 Strong Curvature: The 90° Bend Case

The second benchmark evaluated a tighter 90° pipe bend characterized by a severe curvature ratio of $R_c = 2.8D$. Despite operating at a very similar bulk Reynolds number of $Re_D \approx 43,000$ compared to the U-bend, this significantly tighter bend radius pushed the Dean number up to $De \approx 18,000$. Because the bulk flow speed and Reynolds number remain effectively constant between these two benchmark cases, this comparison successfully isolates the geometric effect of streamline curvature.

As shown in Figure 2.4, examining the quantitative velocity profile at the bend exit ($\theta = 90^\circ$) clearly demonstrates the breakdown of the SA model. Driven primarily by the tighter bend radius and amplified anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall ($y/D = 0$) when compared against both the $k - \omega$ SST prediction and the experimental data from Enayet et al. [18]. This failure is also visually apparent when comparing the side-by-side velocity contours.


(A) $k - \omega$ SST Contour

(B) Spalart-Allmaras Contour



(c) Velocity Profiles

FIGURE 2.3: Flow predictions for a 180° U-bend ($R_c \approx 4.46D$, $Re_D \approx 44,700$, $De \approx 15,000$). The SA model accurately matches the higher-fidelity $k - \omega$ SST contours and directly aligns with the experimental data from Takamasa and Kondo [41].

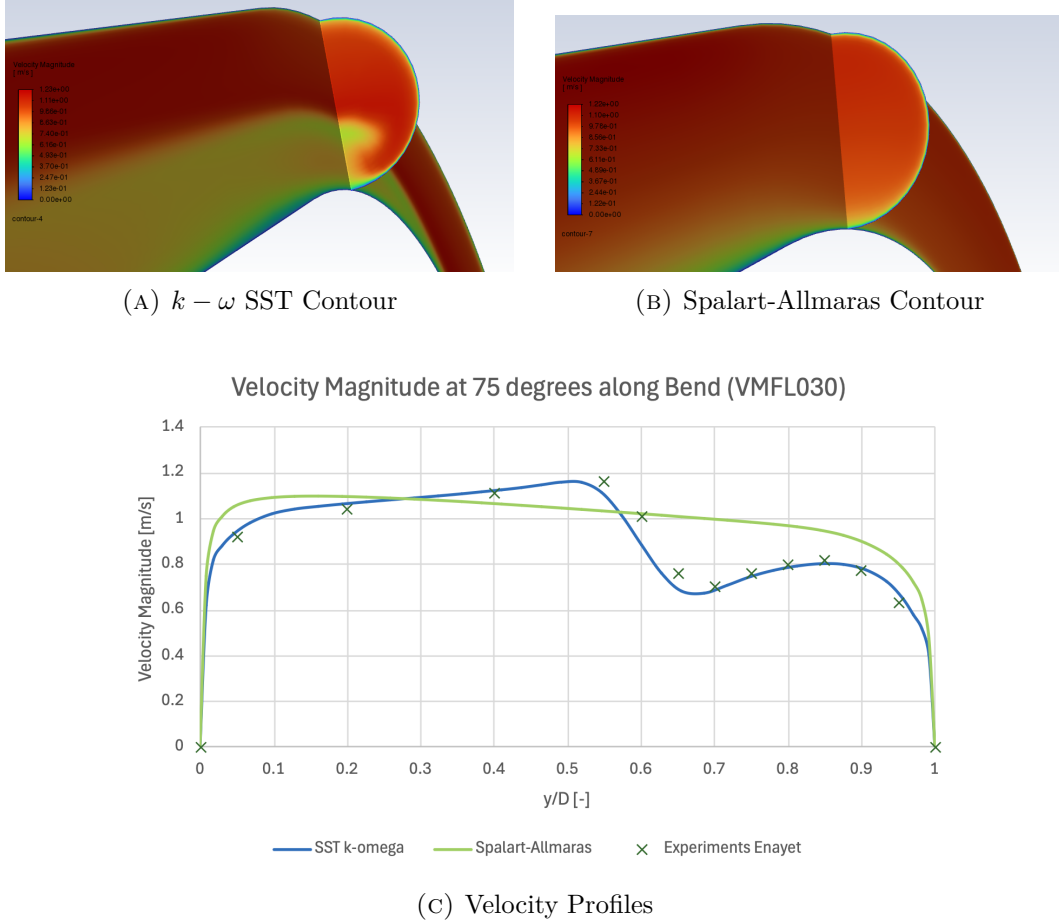


FIGURE 2.4: Flow predictions at the 90° pipe bend exit ($R_c = 2.8D$, $Re_D \approx 43,000$, $De \approx 18,000$). The SA model over-smooths the velocity gradients near the inner wall, deviating significantly from both the $k - \omega$ SST model and the experimental data from Enayet et al. [18].

2.7 Conclusion

This chapter directly addresses the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model. The theoretical breakdown and Ansys benchmark studies confirm that for internal geometries with moderate curvature (up to $De \approx 15,000$), the SA model performs accurately and successfully captures the primary flow physics. Furthermore, because it is a wall-resolved model, it avoids the numerical pitfalls of empirical wall functions, making it exceptionally well-suited for boundary-evolving topology optimization. However, its physical limitation lies in the isotropic Boussinesq assumption. When streamline curvature induces strong anisotropic secondary flows (heuristically observed to cause model breakdown in this case beyond a threshold of $De \approx 18,000$), the uncompensated SA model systematically overpredicts eddy viscosity and fails to capture flow separation. Consequently, while the SA model provides the necessary numerical stability to drive the optimization algorithms in FEniCS, the final topological designs must be critically verified. Any complex manifold generated by the SA optimizer must be subsequently evaluated in commercial software using a higher-fidelity two-equation model to confirm its true aerodynamic performance.

Chapter 3

Implementation and Verification of the Spalart-Allmaras Model in FEniCS

3.1 Introduction to FEniCS and the Finite Element Method

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working computational solver. The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM) and provides a "black box" graphical interface, FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [29].

To solve a PDE in FEniCS, the governing equations (the "strong form") must be mathematically transformed into a "weak formulation" (or variational form). FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [27]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it allows for the automated derivation of the adjoint equations required for gradient-based optimization algorithms.

To ensure complete transparency and reproducibility, the entire custom codebase developed for this thesis is available in the public GitHub repository `TurbulentT0`. To maintain modularity, the repository is divided into two primary directories. The `TurbulenceModels` directory contains the standalone Spalart-Allmaras fluid dynamics engine and the verification benchmarks discussed throughout this chapter. Conversely, the `FluidT0` directory houses the optimization algorithms, penalty formulations, and objective functions that will be detailed in Chapters 4 and 5.

3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in } \Omega \quad (3.1)$$

where p is the pressure field, f is a source term (derived from the velocity divergence), and Ω is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function v and integrated over the entire domain Ω , following standard Galerkin finite element procedures [29, 27]:

$$-\int_{\Omega} (\nabla^2 p) v \, dx = \int_{\Omega} f v \, dx \quad (3.2)$$

Applying integration by parts (Green's first identity) to the left-hand side yields:

$$\int_{\Omega} \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_{\Omega} f v \, dx \quad (3.3)$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation fundamentally lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Furthermore, Neumann boundary conditions (e.g., a prescribed pressure gradient at an outlet) can be naturally substituted directly into the boundary integral $\int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds$. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

3.2 Numerical Solution Strategy for Fluid Dynamics

Solving the full Reynolds-Averaged Navier-Stokes (RANS) equations is numerically challenging due to the non-linear convective terms, the saddle-point nature of the coupled velocity-pressure system, and the intense cross-coupling with the turbulence model. To achieve a stable and efficient steady-state solution in FEniCS that seamlessly translates into the topology optimization framework, the governing equations are rigorously decoupled.

3.2.1 Decoupling the Physics: The Picard Iteration Scheme

Because the momentum equation relies on the turbulent eddy viscosity (ν_t), and the turbulence transport equation relies on the convective velocity ($\mathbf{u}$), solving the entire fluid-turbulence system simultaneously in a massive monolithic matrix frequently leads to ill-conditioning and solver divergence.

To resolve this, the forward physics are decoupled using a **Picard iteration scheme** (successive substitution), a robust approach for segregated RANS solvers

[20, 44]. During each Picard step, the flow solver advances using the eddy viscosity from the previous iteration. The newly updated velocity field is then passed to the Spalart-Allmaras solver to calculate a new eddy viscosity field. This alternating sequence forms the outer loop of the solver, iteratively updating the decoupled equations until the fully coupled physical system reaches a steady-state equilibrium.

3.2.2 Incremental Pressure Correction Scheme (IPCS)

Within the Picard loop, the Navier-Stokes equations themselves must be solved. Rather than solving for velocity and pressure simultaneously, this implementation utilizes an Incremental Pressure Correction Scheme (IPCS). This algorithmic splitting is fundamentally based on the fractional-step projection methods originally pioneered by Chorin [15] and Temam [42]. To nurse the non-linear convective momentum terms to a stable steady state, a pseudo-time stepping approach (Δt) is employed strictly for the flow variables. The IPCS algorithm splits the fluid equations into three sequential steps per pseudo-time step:

1. **Tentative velocity step:** The momentum equation is solved using the pressure field from the previous iteration (p^n) to compute an intermediate, non-divergence-free velocity field ($\mathbf{u}^*$). Because this is a RANS framework, the diffusion term utilizes the effective kinematic viscosity, $(\nu + \nu_t)$. Treating the convective term explicitly and the diffusive term implicitly, the semi-discrete equation is:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla) \mathbf{u}^* - \nabla \cdot ((\nu + \nu_t^n) \nabla \mathbf{u}^*) = -\nabla p^n \quad (3.4)$$

2. **Pressure correction step:** A Poisson equation is solved to find the pressure variation that enforces the continuity constraint ($\nabla \cdot \mathbf{u}^{n+1} = 0$). The equation for the new pressure field (p^{n+1}) is derived as:

$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^* \quad (3.5)$$

3. **Velocity update step:** The tentative velocity is strictly corrected using the gradient of the pressure difference to yield a mass-conserving velocity field ($\mathbf{u}^{n+1}$) for the current iteration:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n) \quad (3.6)$$

The simulation is deemed fully converged to a steady state when the L_2 norm of the residual difference between successive Picard iterations falls below a defined tolerance limit (ϵ_{tol}) for both the fluid and turbulence fields.

3.3 Detailed Formulation of the Spalart-Allmaras Model

As concluded in Chapter 2, the Spalart-Allmaras (SA) one-equation model was selected for its balance of physical accuracy and numerical stability. Unlike the transient flow equations, the SA transport equation is solved in its purely steady-state form within the Picard loop. The steady transport equation for the continuous nonlinear working variable $\tilde{\nu}$ is given by:

$$\mathbf{u} \cdot \nabla \tilde{\nu} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y} \right)^2 + \frac{1}{\sigma} [\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2] \quad (3.7)$$

where ν is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, c_{b1} and c_{w1} are calibrated closure constants, $\tilde{S}$ is a modified vorticity magnitude, f_w is the wall-damping function, and y is the physical distance to the nearest solid boundary [37].

3.3.1 Closure Terms and Calibration Constants

The Spalart-Allmaras model relies on several interconnected closure functions to accurately capture boundary layer physics. The production term relies on a modified vorticity magnitude, $\tilde{S}$, which prevents the destruction of turbulence in regions where the vorticity goes to zero, while maintaining correct scaling near the wall [37]. It is defined as:

$$\tilde{S} = S + \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2} \quad (3.8)$$

where $S = \sqrt{2\Omega_{ij}\Omega_{ij}}$ is the scalar magnitude of the vorticity tensor, κ is the von Kármán constant, and y is the distance to the nearest wall. The function f_{v2} is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \quad (3.9)$$

Similarly, the wall destruction term is modulated by the non-dimensional function f_w , which accelerates the decay of turbulent viscosity inside the viscous sublayer:

$$f_w = g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6} \quad (3.10)$$

where the intermediate variables g and r are evaluated as:

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min \left(\frac{\tilde{\nu}}{\tilde{S} \kappa^2 y^2}, 10 \right) \quad (3.11)$$

The upper bound of 10 on r ensures numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

It should be explicitly noted that the original formulation by Spalart and Allmaras [37] includes additional geometric trip functions (f_{t1} and f_{t2}) designed to manually trigger the transition from laminar to turbulent flow at specific user-defined coordinates. However, following the topology optimization methodology established by Yoon [48], these transition functions are deliberately omitted from this thesis. Because the topological boundaries are unknown *a priori* and evolve during optimization, tracking discrete trip points is mathematically unfeasible; therefore, the model assumes a fully turbulent flow regime.

To close the system, the standard empirical calibration constants formulated by Spalart and Allmaras are utilized throughout the FEniCS implementation [34]. Notably, the von Kármán constant is set to 0.41 to perfectly align with the standard OpenFOAM and generalized industrial implementations[cite: 3]:

$$\begin{aligned} c_{b1} &= 0.1355 & \sigma &= 2/3 & c_{b2} &= 0.622 \\ \kappa &= 0.41 & c_{w2} &= 0.3 & c_{w3} &= 2.0 \\ c_{v1} &= 7.1 & c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma} \end{aligned}$$

3.3.2 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance y . In a standard static CFD simulation, y is purely geometric and can be pre-computed. However, to ensure this verification solver translates seamlessly into the topology optimization code (where solid boundaries evolve continuously), the implementation abandons discrete geometric searches in favor of a PDE-based approach.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability—a requirement for later adjoint derivations—a relaxed Eikonal equation proposed by Yoon [48] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G (\nabla \cdot \nabla G) = (1 + 2\sigma_w) G^4 \quad (3.12)$$

Here, G is an intermediate, non-dimensional field variable. The parameter σ_w is a relaxation constant (typically set to 0.1) that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where G_0 is an arbitrarily large constant (e.g., $G_0 = 20.0$). Once the field G is solved over the domain, the physical wall distance y is recovered algebraically at every node using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0} \quad (3.13)$$

This system is implemented in the core solver suite within the `Utilities.py` script. As shown in Listing 3.1, once the non-linear variational problem for G is solved, the physical distance y is recovered and strictly bounded from below to prevent numerical singularities.

LISTING 3.1: Implementation of Yoon’s relaxed Eikonal equation for wall-distance calculation, extracted from `Utilities.py`.

```

# Weak form corresponding to Yoon 2016 Eq. (19):
F = (
    (Constant(1.0) - sigma_w_const) * inner(grad(G), grad(G)) * z
    * dx
    - sigma_w_const * G * inner(grad(G), grad(z)) * dx
    - (Constant(1.0) + Constant(2.0) * sigma_w_const) * G**4 * z *
    dx
)
problem = NonlinearVariationalProblem(F, G, bcs=wall_bcs, J=
    derivative(F, G))
solver = NonlinearVariationalSolver(problem)
solver.solve()

# Bound G to prevent division by zero
G.vector()[:] = np.maximum(G.vector()[:], float(g_floor))

# Recover physical wall distance (y)
y = project(Constant(1.0) / G - Constant(1.0) / g0_const,
    Space)
return bound_from_bellow(y, 0.0)

```

3.4 Variational Formulation and FEniCS Implementation

The standard Galerkin weak formulation is constructed by multiplying the steady transport Equation 3.7 by a test function ξ and applying integration by parts to the diffusive terms. Because the equation is highly non-linear, robust convergence is dependent on the stability of the external velocity field provided by the Picard outer loop.

To manage this non-linearity, the equation is linearized within the Picard iteration scheme by freezing certain parameters using the most recent value of the working variable, $\tilde{\nu}^0$. Specifically, the turbulence production and cross-diffusion terms are treated as explicit sources evaluated using $\tilde{\nu}^0$ [cite: 3]. Conversely, the wall destruction term is linearized into an implicit reaction sink by scaling the continuous $\tilde{\nu}$ against the frozen variable, taking the form $c_{w1}f_w(\tilde{\nu}^0/y^2)\tilde{\nu}$ [cite: 3].

The explicit translation of this mathematical formulation is handled by the `SpalartAllmarasSteadyState` class within the `TurbulenceModel_SpalartAllmaras.py` script[cite: 3]. As demonstrated in Listing 3.2, the Unified Form Language (UFL) mirrors the analytical weak form utilizing the most recent convective velocity field (`external_velocity`) passed from the IPCS flow solver.

LISTING 3.2: FEniCS implementation of the steady-state SA standard Galerkin weak formulation. Python dictionary lookups have been simplified for readability.

```

def construct_forms(self, external_velocity):
    self._construct_turbulent_quantities(external_velocity)
    sigma = Constant(2.0 / 3.0)

```

```
# Steady-state SA transport equation weak form
FNT = (
    dot(external_velocity, nabla_grad(self._nu_tilde)) * self._xi
    * self._dx
    + inner((self._nu + self._nu_tilde0) / sigma * grad(self.
        _nu_tilde), grad(self._xi)) * self._dx
    + self._react_nt * self._nu_tilde * self._xi * self._dx
    - self._source_nt * self._xi * self._dx
)

self._a_nt = lhs(FNT)
self._l_nt = rhs(FNT)
```

3.5 General Solver Configuration and Numerical Parameters

Before detailing the specific verification benchmarks, it is necessary to establish the general numerical configuration of the FEniCS solver. Both the channel flow and the U-bend verifications were simulated using the segregated algorithmic structure detailed earlier: an outer Picard fixed-point loop couples the fluid momentum equations to the Spalart-Allmaras transport equation, while an inner IPCS resolves the velocity-pressure subsystem via pseudo-time stepping.

Because the U-bend domain introduces severe geometric curvature, strong convective transport, and adverse pressure gradients, it represents a significantly more demanding numerical challenge than a fully developed, unidirectional channel flow. Consequently, the U-bend simulations necessitated specific numerical controls to maintain stability, including a heavily refined IPCS pseudo-time step ($\Delta t = 1.0 \times 10^{-4}$ s), a maximum of 400 coupled Picard iterations, up to 800 IPCS steps per Picard update, a pressure relaxation factor of 1.0, and tighter under-relaxation of the turbulence field updates ($\alpha_{\tilde{\nu}} = 0.01$) [cite: 2]. To ensure reproducibility, the comprehensive matrices detailing all physical boundary conditions, geometric dimensions, and exact numerical solver parameters for both benchmarks have been compiled and are documented extensively in Sections A.1 and A.2 of Appendix A (specifically in Tables A.2 and A.6).

3.6 Verification Benchmark 1: Channel Flow

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate and verify turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom bound-

aries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and utilizing the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20,300$. By explicitly defining the Reynolds number based on the channel height rather than the mathematical hydraulic diameter, this formulation directly aligns with academic conventions for fundamental 2D channel flow verification. The complete definition of the geometric, physical, and boundary-condition parameters can be found in Table A.1 of Appendix A.

3.6.1 Mesh Resolution Requirements

Because the SA formulation is a "low-Reynolds" model that does not utilize empirical wall functions, it requires mathematical integration directly through the viscous sublayer. To satisfy this strict boundary layer requirement, a custom structured inflation mesh was generated for the channel flow. Because the domain is a simple periodic rectangle, this highly anisotropic grid was constructed programmatically utilizing NumPy arrays and converted into a FEniCS-compatible format via the `meshio` library. Based on the target bulk Reynolds number, this approach enforced a physical first-cell height of approximately 1.7×10^{-3} meters, aiming for a non-dimensional wall distance of $y^+ \approx 1$ across the entire domain.

Furthermore, to verify that the numerical results were not heavily influenced by spatial discretization, a formal grid independence study was conducted. The full results of this spatial verification are documented in Appendix A.1.3. Specifically, the geometric properties of the mesh hierarchy are detailed in Table A.3, and visual confirmation of asymptotic convergence is provided by the velocity profiles in Figure A.1.

3.6.2 FEniCS vs. Ansys Comparison

Prior to this comparative analysis, the Ansys baseline itself was verified for grid independence to ensure a robust standard (see Table A.4 and Figure A.2 in Appendix A). The channel flow was then simulated in FEniCS using the decoupled Picard iteration method and compared against the Ansys Fluent baseline.

As shown in Figure 3.1, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. Furthermore, the quantitative line plots confirm that the velocity profiles generated by the custom FEniCS model closely track the Ansys reference data. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center. To provide a rigorous, objective comparison, the quantitative errors and macroscopic metrics are summarized in Table 3.1.

This strong alignment supports the verification that the core Navier-Stokes solver, the relaxed Eikonal field, and the SA production/destruction terms are functioning as intended before introducing the geometric complexities of the U-bend.

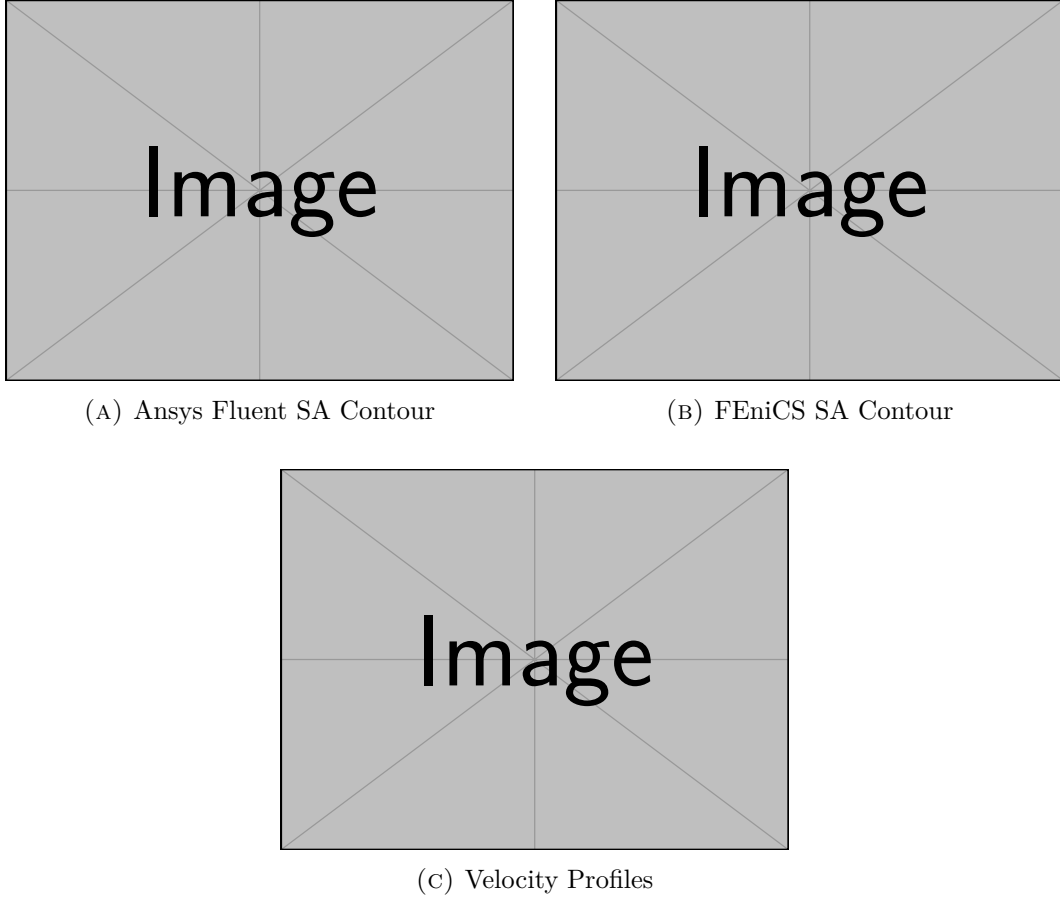


FIGURE 3.1: Verification of the 2D Channel Flow ($Re_H \approx 20, 300$). Comparison of the qualitative velocity contours between the Ansys Fluent baseline (a) and the custom FEniCS implementation (b), alongside the quantitative velocity profiles extracted across the channel (c).

TABLE 3.1: Quantitative comparison of flow metrics and extracted errors for the 2D channel benchmark.

Metric	Ansys Fluent	Custom FEniCS	Relative Error / Difference
Bulk Velocity (U_{bulk}) [m/s]	[TBD]	[TBD]	[TBD]%
Friction Coefficient (C_f)	[TBD]	[TBD]	[TBD]%
Max Pointwise Velocity Error	–	–	[TBD]%
Relative L_2 Velocity Error	–	–	[TBD]%

3.7 Verification Benchmark 2: U-Bend Geometry

Following the channel flow verification, the solver was tested on the highly convective separating flows of the U-bend geometry evaluated in Chapter 2.

As discussed in Appendix A.2.3, the verification performed here utilized a mod-

ified computational domain to ensure efficiency. First, the domain was reduced to a 2D geometry corresponding directly to the symmetry mid-plane of the full 3D case. Since the FEniCS domain is mathematically a 2D planar approximation, it inherently cannot resolve true 3D Dean secondary vortices. Instead, this benchmark serves to validate the 2D symmetry-plane approximation and the solver’s ability to capture curvature-induced separation and momentum redistribution. Second, to reduce unnecessary computational overhead during these verification sweeps, the extended straight inlet and outlet sections (1.555 m) of the standard benchmark were truncated to $30D$ (0.840 m). The mathematical justification for this truncation is plotted in Figure A.3 of Appendix A.2.3.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42$ m/s) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m²/s), while a zero relative static pressure ($p = 0$ Pa) was assigned to the outlet. While the domain is 2D, the 3D physical pipe diameter ($D = 0.028$ m) and geometric radius of curvature ($R_c = 0.125$ m) were retained as the characteristic length scales. Conserving these parameters ensures that the computed Reynolds number ($Re_D \approx 44,700$) and Dean number ($De \approx 15,000$) match the 3D Ansys reference case. For an overview of the physical setup and boundary conditions applied to this geometry, the reader is referred to Table A.5 in Appendix A.

3.7.1 Mesh Resolution Requirements

Because this formulation utilizes a standard Galerkin weak form without spatial upwind stabilization, numerical stability in highly convective flows relies on adequate grid resolution.

The U-bend geometry features curved inner and outer arcs, complex corner transitions, and anisotropic boundary-layer inflation around non-orthogonal walls. To manage this geometric complexity while satisfying the $y^+ \approx 1$ resolution requirement, a custom mesh was generated using **Gmsh** [22]. The first element height adjacent to the wall was controlled to 1.20×10^{-5} meters across the entire domain. As documented in Appendix A.2.4, independence studies were conducted. The inflation parameters and element densities of the Gmsh hierarchy are listed in Table A.7, and the extracted velocity profiles confirming grid convergence at the bend exit are illustrated in Figure A.4.

3.7.2 FEniCS vs. Ansys Comparison

Having validated the spatial resolution of the FEniCS mesh, an equivalent grid independence study was simultaneously executed for the commercial Ansys Fluent baseline (detailed in Table A.8 and Figure A.5 of Appendix A.2.4). With both solvers operating on spatially converged grids, an independent Ansys Fluent simulation was executed on the shortened 2D domain.

As shown in Figure 3.2, the global flow fields and flow separation regions observed in the qualitative velocity contours demonstrate visual agreement. To provide a more robust assessment, velocity data was extracted along a cross-sectional line probe

3. IMPLEMENTATION AND VERIFICATION OF THE SPALART-ALLMARAS MODEL IN FEniCS

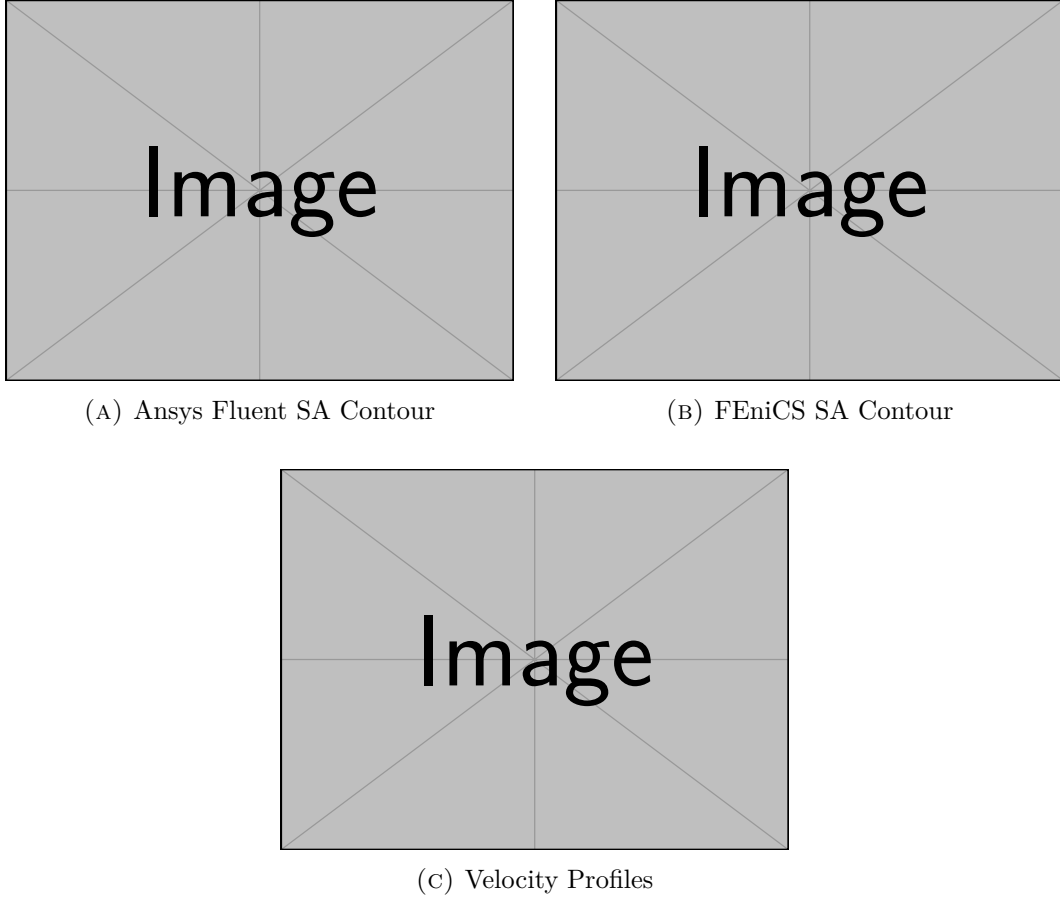


FIGURE 3.2: Verification of the 2D U-Bend Flow ($Re_D \approx 44,700$, $De \approx 15,000$). Comparison of the qualitative velocity contours between the Ansys Fluent baseline (a) and the custom FEniCS implementation (b), showing separation characteristics, alongside the quantitative velocity profiles extracted at the bend exit (c).

positioned directly at the bend exit. The quantitative velocity profile computed by the custom FEniCS implementation traces the commercial Ansys results closely, adequately capturing the boundary layer growth and the momentum shift toward the outer wall.

Table 3.2 summarizes the quantitative error margins and the macroscopic pressure drop (Δp) across the 2D U-bend domain for both solvers.

TABLE 3.2: Quantitative comparison of flow metrics and extracted errors for the 2D U-bend benchmark.

Metric	Ansys Fluent	Custom FEniCS	Relative Error / Difference
Pressure Drop (Δp) [Pa]	[TBD]	[TBD]	[TBD]%
Max Pointwise Velocity Error	–	–	[TBD]%
Relative L_2 Velocity Error	–	–	[TBD]%

This general agreement in both localized velocity gradients and global energy dissipation indicates that the framework provides a credible foundation for the pressure-drop minimization objectives in the subsequent topology optimization campaigns.

3.8 Conclusion

This chapter has detailed the mathematical formulation, stabilization strategy, and verification of the steady-state Spalart-Allmaras model within the FEniCS framework. By evaluating the solver against Ansys Fluent benchmarks, this implementation addresses the second research question of this thesis: achieving numerical stability and establishing reasonable quantitative agreement in a custom finite element solver to support topology optimization routines.

First, numerical stability was achieved without the need for complex mathematical stabilization artifacts. By decoupling the Navier-Stokes and Spalart-Allmaras equations via a Picard iteration scheme, and utilizing an Incremental Pressure Correction Scheme (IPCS) to manage the convective fluid terms, the highly non-linear system was driven to equilibrium. Supported by high-density mesh resolution and Yoon’s relaxed Eikonal equation, this standard Galerkin formulation effectively mitigated numerical oscillations.

Second, quantitative agreement was assessed. The solver produced velocity profiles that showed good agreement with the commercial Ansys Fluent baseline in simple periodic channel flows ($Re_H \approx 20,300$) and adequately captured the 2D boundary layer features in a curved U-bend ($Re_D \approx 44,700$, $De \approx 15,000$). While approximations inherent to 2D domains limit the direct translation to true 3D Dean vortices, this decoupled Picard FEniCS solver demonstrates sufficient accuracy in capturing curvature-induced momentum shifts. This suggests that the implementation is suitable for driving the continuous adjoint topology optimization campaigns presented in Chapters 4 and 5.

Chapter 4

Topology Optimization for Turbulent Flows: Methodology

4.1 Introduction

Building upon the verified Spalart-Allmaras (SA) fluid dynamics solver [36, 37] developed in Chapter 3, this chapter introduces the mathematical framework required to perform Topology Optimization (TO) in the turbulent regime. The core objective of this methodology is to algorithmically determine the optimal distribution of fluid channels within a given design domain to reduce a prescribed hydraulic loss metric, such as power dissipation or pressure loss [3].

This chapter details the density-based fictitious-domain formulation used to represent solid and fluid regions on a fixed computational mesh [13, 21]. It then explains the specific turbulent-flow ingredients retained for the remainder of the thesis: a Brinkman-type momentum resistance, a power-law topology penalty for the SA and wall-distance equations, and the reciprocal penalized wall-distance formulation used by the Alexandersen pipe-bend computations.

To solve this optimization problem in a computationally tractable manner, continuous adjoint sensitivity analysis is required [32, 31]. The optimization results reported in this thesis use the frozen-turbulence, or Constant Eddy Viscosity (CEV), assumption [47, 35]. This assumption is stated explicitly because it is both the practical reason the turbulent optimization loop remains stable and the main limitation of the sensitivity model.

Following the derivation of these sensitivities, the continuous optimization problem is formally defined, establishing the admissible objective functions and the strict fluid volume constraints over explicitly defined design regions. Finally, the mathematical techniques for active-domain density filtering [28] and Heaviside projection [45] are detailed alongside their FEniCS unified form language (UFL) implementation [4, 29] to ensure the generation of manufacturable designs. The algorithms, penalty formulations, and execution scripts are housed within the `FluidTO` directory of the public repository, with the frozen-turbulence workflow implemented in `TurbulentTO_Frozen.py`.

4.2 The Fictitious Domain Method and Brinkman Penalization

In standard shape optimization, the solid-fluid boundary is explicitly defined by the mesh, and the nodes are physically moved to improve the design. In density-based topology optimization, however, the computational domain (Ω) remains strictly fixed. To create solid walls and fluid channels within this fixed grid, the Fictitious Domain Method is employed.

Originally applied to Stokes flow by Borrvall and Petersson [13] and extended to the Navier-Stokes equations by Gersborg-Hansen et al. [21], this method assigns a continuous design variable to the spatial coordinates in the domain. In this specific implementation convention, a physical topology value of $\bar{\gamma} = 1$ represents a pure **fluid** channel, while $\bar{\gamma} = 0$ represents a **solid**, impermeable structure. To force the fluid velocity to zero inside the solid regions without mathematically removing the finite elements, the fluid is treated as a fictitious porous medium.

This flow resistance is quantified by an inverse permeability term, or Brinkman penalty function, denoted as $\alpha(\bar{\gamma})$. To ensure the optimization algorithm (which relies on continuous gradients) can transition smoothly between fluid and solid states, α must be a continuous interpolation function. This solver utilizes a convex Rational Approximation of Material Properties (RAMP) interpolation [38] scaled between a maximum solid penalty (α_{solid}) and a minimum fluid penalty (α_{fluid}):

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid}) \frac{\bar{\gamma}(1+q)}{\bar{\gamma}+q} \quad (4.1)$$

where α_{solid} is a sufficiently large penalty factor that completely stagnates the flow when $\bar{\gamma} = 0$, and q is a tuning parameter that controls the convexity of the penalization curve. Unlike traditional structural optimization which relies on the SIMP power-law [12], fluid optimization generally avoids SIMP for momentum because its derivative vanishes at $\bar{\gamma} = 0$. By utilizing the RAMP formulation, a strictly non-zero gradient is guaranteed across the entire design domain.

4.2.1 The Density Chain and Domain Bounding

While conceptually represented by the physical topology field in the governing equations, the computational implementation dictates that the layout is managed through a distinct chain of density representations to manage design and non-design domains. To explicitly avoid mathematical confusion with the fluid mass density (ρ) utilized in the Reynolds-Averaged Navier-Stokes equations, this thesis exclusively utilizes the γ notation for all topological variables. However, these variables correspond directly to the exact implementation code variables as follows:

- **Raw design density** ($\gamma \equiv \rho_{raw}$): The baseline parameter manipulated by the optimizer.
- **Filtered density** ($\tilde{\gamma} \equiv \rho_f$): The spatially smoothed design field.

- **Projected effective density** ($\bar{\gamma} \equiv \rho_{\text{eff}}$): The bounded, physical topology field evaluated by the governing equations.

The raw design density, γ , is defined as a Discontinuous Galerkin (DG0) [24] function over the entire mesh. The optimization algorithm strictly manipulates the degrees of freedom corresponding to the actively optimized design space (**ActiveDV**), while passive (non-design) cells are subsequently restored to their prescribed bounding values.

This raw variable is subjected to a spatial PDE filter [28] to yield a smoothed field $\tilde{\gamma}$. To recover sharp boundaries and strictly preserve required non-design regions, the filtered density is then projected and mapped into the bounded effective density ($\bar{\gamma}$) [23]:

$$\bar{\gamma} = \gamma_{\text{lower}} + (\gamma_{\text{upper}} - \gamma_{\text{lower}})\Pi_{\beta}(\tilde{\gamma}) \quad (4.2)$$

where Π_{β} is the projection operator, and γ_{lower} and γ_{upper} define the absolute permissible bounds for any given cell. This rigorous bounding guarantees that predefined features, such as non-design fluid inlet extensions ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 1$) or explicitly constrained structural walls ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 0$), remain rigidly fixed without interfering with the mathematical behavior of the active optimization space.

The computational mesh is therefore partitioned into an active design domain and passive non-design domains. The active design domain contains the cells whose raw densities are updated by MMA. Fluid non-design domains are fixed as permanent fluid, typically to preserve inlet and outlet extensions, buffer channels, or measurement regions. Solid non-design domains are fixed as permanent solid, typically to impose external walls, blocked regions, or geometric features that must not be modified. These passive cells still participate in the forward flow, turbulence, and wall-distance solves, but they are excluded from the optimization masks: the volume constraint is evaluated only over the configured volume mask Ω_V , and domain-integral objectives are evaluated only over the configured objective mask Ω_D . Consequently, prescribed inlet/outlet buffers and mandatory solid walls do not count toward the optimized fluid fraction and do not directly bias the domain-integrated objective.

4.3 Penalization of the Governing Equations

To effectively optimize turbulent manifolds, the respective penalties must be rigorously integrated into the turbulence model and the wall-distance calculations to ensure the physics remain consistent as the solid boundaries evolve. All physical penalties strictly utilize the projected effective density, $\bar{\gamma}$.

4.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The primary application of the Brinkman penalty is the modification of the steady-state, incompressible RANS momentum equation. The RAMP-interpolated inverse permeability $\alpha(\bar{\gamma})$ is added as a linear sink term (Darcy friction) to the momentum

conservation equation:

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] - \alpha(\bar{\gamma}) \mathbf{u} \quad (4.3)$$

It is critical to note the retention of the full symmetric rate-of-strain tensor. Here, ρ strictly denotes the fluid mass density. Because the turbulent eddy viscosity (μ_t) varies drastically across the spatial domain [33], the effective fluid viscosity is no longer constant, and the transposed term exerts a non-zero physical influence on the momentum balance. In regions where the optimizer places solid material ($\bar{\gamma} \rightarrow 0$), the enormous magnitude of $\alpha_{solid} \mathbf{u}$ dominates all inertial and viscous forces, forcing the local velocity mathematically to zero.

4.3.2 Modified Spalart-Allmaras Transport Equation

As the Fictitious Domain Method forces velocity to zero in the solid regions, the turbulence generation inside those regions should theoretically cease. However, due to numerical diffusion, turbulent eddy viscosity ($\tilde{\nu}$) can inadvertently bleed into the solid regions. This unphysical behavior destabilizes the solver and heavily distorts the adjoint sensitivity analysis.

To strictly enforce a laminar, zero-turbulence state inside the solid structures, a corresponding penalty sink term is added to the Spalart-Allmaras transport equation [48, 17]:

$$\mathbf{u} \cdot \nabla \tilde{\nu} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y} \right)^2 + \frac{1}{\sigma} [\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} (\nabla \tilde{\nu})^2] - R_{\tilde{\nu}}(\bar{\gamma}) \tilde{\nu} \quad (4.4)$$

Here $R_{\tilde{\nu}}(\bar{\gamma})$ is an artificial topology-dependent reaction coefficient. It is not part of the original SA model; it is introduced only to suppress spurious values of $\tilde{\nu}$ in the fictitious solid and in intermediate porous material.

4.3.3 Power-Law Topology Penalty

The SA and wall-distance penalties require an interpolation from the physical density field $\bar{\gamma}$ to a solid indicator. For the computations emphasized in this thesis, this interpolation is the positive-part power-law indicator

$$S_{\text{pow}}(\bar{\gamma}; N) = [\max(1 - \bar{\gamma}, 0)]^N, \quad (4.5)$$

so that a generic reaction penalty is written as $R_{\psi}(\bar{\gamma}) = \alpha_{\psi} S_{\text{pow}}(\bar{\gamma}; N_{\psi})$, with $\psi \in \{\tilde{\nu}, G\}$ depending on the equation being penalized. The indicator is zero in pure fluid and increases smoothly toward one as the density approaches solid. For thresholded wall-distance updates, the same construction is applied to a shifted solid indicator,

$$S_{\text{pow},\theta}(\bar{\gamma}; N, \theta_s) = \left[\max \left(\frac{\theta_s - \bar{\gamma}}{\theta_s}, 0 \right) \right]^N. \quad (4.6)$$

4.3.4 Reciprocal Penalized Wall-Distance Equation

The Spalart-Allmaras model is highly sensitive to the wall distance, y . While differential equations for wall distance computation are well-established [43], topology optimization introduces a unique challenge. The location of the wall is not a static mesh boundary, but rather the dynamically evolving $\bar{\gamma} = 0$ contour. Therefore, the relaxed Eikonal equation must be penalized to dynamically recognize the fictitious solid as a physical boundary [48]:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) = (1 + 2\sigma_w)G^4 + R_G(\bar{\gamma})(G - G_0) \quad (4.7)$$

Consistent with the SA transport modifications, the reciprocal wall-distance equation uses a staged multiplier paired with the power-law solid indicator. In the Alexandersen pipe-bend configuration, $R_G(\bar{\gamma}) = \alpha_G S_{\text{pow},\theta}(\bar{\gamma}; N_G, \theta_s)$ with a design-driven wall-density source. This is the implementation mode denoted `reciprocal_penalized`. The algebraic recovery function

$$y = \frac{1}{G} - \frac{1}{G_0} \quad (4.8)$$

then collapses the SA wall distance toward zero inside topology-created solid regions while retaining the physical wall distance in fluid regions.

4.4 Boundary Conditions and Turbulence Initialization

To stabilize the topology optimization and ensure physical accuracy, boundary constraints are strictly managed across both fluid and turbulence regimes. Non-design fluid buffer regions are strictly mandated at inlets and outlets to isolate the active optimization domain from artificial numerical shear. The framework robustly handles boundary interactions by incorporating pressure outlets coupled with optional velocity-component constraints, as well as pressure pin controls to fully constrain the discrete PDE problem.

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

4.5 Sensitivity Analysis and the Frozen Turbulence Assumption

With the forward state equations fully defined and penalized, the topology optimization algorithm requires the gradient (sensitivity) of the objective function with

respect to the design variable. Because evaluating finite differences for every mesh element is computationally impossible, the continuous adjoint method is employed [31].

However, calculating the fully coupled, exact adjoint for turbulent flows is mathematically formidable. As demonstrated by Zymaris et al. [52], formulating the exact continuous adjoint for the Spalart-Allmaras model requires differentiating the highly non-linear production and destruction terms, accounting for the variations in the distance function, and managing the intense cross-coupling with the momentum equations. The complexity of these exact adjoints frequently introduces severe numerical stiffness and divergence.

To make the gradient calculation both numerically robust and computationally feasible, the primary optimization workflows in this implementation rely on the **Frozen Turbulence Assumption**, frequently referred to in the literature as the Constant Eddy Viscosity (CEV) assumption [47, 35].

4.5.1 Mechanism of Frozen Turbulence

The frozen turbulence assumption simplifies the sensitivity analysis by decoupling the turbulence transport from the design update. During the adjoint formulation, the turbulent eddy viscosity field (ν_t) generated by the forward SA solver is assumed to be locally independent of infinitesimally small changes in the physical topology ($\bar{\gamma}$). Mathematically, this implies that during the calculation of the adjoint sensitivities:

$$\frac{\partial \nu_t}{\partial \bar{\gamma}} \approx 0 \quad (4.9)$$

By treating the eddy viscosity field as frozen (a spatially varying but constant parameter), the complex adjoints of the Spalart-Allmaras and Eikonal equations do not need to be solved. The optimizer only solves the standard fluid adjoint equations (utilizing the effective viscosity $\mu + \mu_t$) to determine how a change in topology affects the viscous dissipation.

4.5.2 Justification and Limitations

The primary advantage of the frozen turbulence approach is not that it is universally better, but that it makes the optimization problem solvable within a reasonable numerical and implementation budget. Relative to a fully coupled turbulent adjoint, it offers greater numerical stability, a smaller implementation burden, and a materially lower computational cost per design iteration. This is why frozen turbulence remains common in industrial adjoint workflows and why it is used as the baseline in this thesis [31, 35].

The price of this robustness is that the gradients are approximate reduced derivatives. Because the algorithm ignores how an infinitesimal topology change would alter future turbulence production, the sensitivity can be inaccurate when the objective is strongly controlled by separated shear layers, stall-like behavior, or other regimes where the eddy viscosity changes sharply with geometry. Wu et al. [47]

show that the CEV assumption can still achieve useful optimization goals in benign flows, but its sensitivity accuracy and adjoint convergence can deteriorate near stall and in unsteady settings. Zymaris et al. [52] further demonstrate, for SA-based continuous adjoints, that including the turbulence-equation adjoint improves derivative accuracy for loss functionals. The present thesis therefore defends the frozen assumption as a practical and verifiable approximation, not as an exact turbulence sensitivity model. Attempts to include the SA turbulence equation in the adjoint were explored during the implementation phase, but the resulting solves were not sufficiently stable for the thesis optimization campaign.

4.6 Optimization Problem Formulation

Using the frozen-turbulence sensitivities described above, the topology optimization task is formulated as a mathematical programming problem, where the optimizer must find a spatial distribution that minimizes a specific performance metric subject to physical and geometric constraints.

4.6.1 Objective Functions

For fluid manifolds, the primary engineering goal is to guide the fluid from the inlets to the outlets with the minimum possible hydraulic loss. In incompressible flow, this loss can be expressed either through a domain-integrated power dissipation objective or through a pressure-based objective, depending on the benchmark configuration and the quantity reported in the reference study.

For dissipation-driven cases, it is crucial to explicitly distinguish the algorithmic optimization objective (J_D) from the true physical validation metrics analyzed later in Chapter 5. The dissipation objective includes artificial Brinkman drag and is integrated exclusively over the configured objective mask (Ω_D):

$$J_D(\mathbf{u}, \bar{\gamma}) = \int_{\Omega_D} \left[\frac{1}{2}(\mu + \mu_t) (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) : (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) + \alpha(\bar{\gamma}) \mathbf{u} \cdot \mathbf{u} \right] d\Omega \quad (4.10)$$

The first term in the integral represents the physical dissipation caused by the effective fluid viscosity. The second term represents the mathematical energy lost when the fluid attempts to penetrate the fictitious solid regions.

For pressure-driven benchmarks with prescribed inflow and fixed outlet pressure, the optimization objective can instead be defined as the average inlet pressure,

$$J_p(p) = \frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p d\Gamma. \quad (4.11)$$

When the outlet pressure is used as the gauge reference, minimizing J_p is equivalent to minimizing the pressure loss required to drive the imposed flow rate through the topology. Unlike Eq. 4.10, this is a boundary integral rather than a volume integral over Ω_D . Its sensitivity is nevertheless mapped back only to the active design degrees of freedom through the same active-mask filter and transpose-filter machinery, so

the prescribed non-design inlet region is used only to measure the pressure and is not itself optimized.

4.6.2 Volume Constraint

Because the goal is to design a targeted manifold or pipe system, a strict volume constraint must be imposed. The volume fraction constraint is enforced via a discrete volume region mask (Ω_V), strictly dictating the proportion of material evaluated by the optimizer:

$$g(\bar{\gamma}) = \frac{\int_{\Omega_V} \bar{\gamma} d\Omega}{V_{\text{mask}}} - V_{\text{max}} \leq 0 \quad (4.12)$$

where V_{mask} is the total volume of the designated integration space, and V_{max} is the maximum allowable fluid volume fraction.

4.6.3 The Complete Continuous Optimization Problem

Combining the selected objective function, the governing equations, and the constraints, the continuous topology optimization problem for turbulent flow is formally stated as:

$$\begin{aligned} & \underset{\gamma(\mathbf{x}) \in [0,1]}{\text{minimize}} && J(\mathbf{u}, p, \bar{\gamma}) \\ & \text{subject to} && \text{Modified RANS Equations (Eq. 4.3)} \\ & && \text{Modified Spalart-Allmaras Equation (Eq. 4.4)} \\ & && \text{Configured Wall-Distance Equation} \\ & && g(\bar{\gamma}) \leq 0 \end{aligned} \quad (4.13)$$

where J denotes the selected benchmark objective, either the dissipation objective J_D in Eq. 4.10 or the inlet-pressure objective J_p in Eq. 4.11. The wall-distance constraint used for the Alexandersen-focused computations is the reciprocal penalized relaxed Eikonal equation in Eq. 4.7, with the wall distance recovered by Eq. 4.8.

To solve this constrained minimization problem, the FEniCS implementation utilizes the Method of Moving Asymptotes (MMA) [40], an optimization algorithm specifically designed for handling large-scale, gradient-based topology optimization problems. MMA approximates the highly non-linear design space using convex asymptotes, relying on strict move limits which cap the maximum allowable change in the active raw design variable (γ) during any single optimization step.

4.7 Density Filtering and Projection

A well-known mathematical complication in density-based topology optimization is the lack of existence of a classical solution. If the design space is left unconstrained, the optimizer will tend to create infinitely thin micro-structures (checkerboarding). To guarantee a mesh-independent and manufacturable design, a minimum length scale must be enforced.

4.7.1 The Discontinuous Galerkin Helmholtz Filter

To prevent checkerboarding, a density filter is applied to the raw design variable generated by the optimizer. Because a continuous nodal definition causes the Brinkman penalty to interpolate smoothly across the interior of boundary elements, creating a semi-permeable gray gradient that allows mass leakage, this implementation discretizes the density field using piecewise constant elements (Discontinuous Galerkin, DG0) [9]. By assigning a single, uniform density value to the entire element, the structural boundaries remain perfectly sharp at the element edges.

Consequently, the standard continuous Helmholtz filter cannot be applied. The filter is mathematically adapted to account for the discontinuous boundaries utilizing an interior penalty method across the element facets (Γ_{int}). Standard volume diffusion is mathematically replaced by interior facet jump penalties:

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} \llbracket \tilde{\gamma} \rrbracket \cdot \llbracket v \rrbracket dS + \int_{\Omega} \tilde{\gamma} v dx = \int_{\Omega} \gamma v dx \quad (4.14)$$

where v is the test function, $\llbracket \cdot \rrbracket$ denotes the jump operator across a facet, h_{avg} is the average cell size, and $r = \frac{R}{2\sqrt{3}}$ is the scaled filter radius.

Active-Mask Normalization

To appropriately handle the severe boundary interactions near designated inlet and outlet buffers, this filtering is strictly active-mask-normalized. Rather than applying the filter blindly across the entire domain, the filtered output evaluates as:

$$\tilde{\gamma} = \frac{H(M\gamma)}{H(M)} \quad (4.15)$$

where H represents the Helmholtz PDE operator and M represents the isolated active design indicator mask. Crucially, during sensitivity analysis, the mathematical transpose of this normalized filter must be explicitly derived and applied to map the calculated objective gradients back into the active DG0 design space. The explicit programmatic integration of this interior penalty filter, active-mask normalization, and the corresponding adjoint transpose logic is demonstrated in Listing 4.1.

LISTING 4.1: Schematic pseudocode of the DG0 Helmholtz filter with active-mask normalization and interior facet jump penalties. Extracted from `TurbulentTO_Frozen.py`.

```
# r is the internal PDE coefficient scaled by 1/(2*sqrt(3))
r = r_filter / (2.0 * 3.0**0.5)

u_filter = TrialFunction(DensitySpace)
v_filter = TestFunction(DensitySpace)
filter_in = Function(DensitySpace)

h_avg = (CellDiameter(mesh)("+" ) + CellDiameter(mesh>("-" )) /
2.0
```

```

def pde_filter_raw(input_field, output_field):
    alpha_dg = 4.0
    helmholtz = (
        r**2 * (alpha_dg / h_avg * dot(jump(v_filter, n), jump(
            u_filter, n))) * dS
        + u_filter * v_filter * dx
        - filter_in * v_filter * dx
    )
    assign(filter_in, input_field)
    solve(lhs(helmholtz) == rhs(helmholtz), output_field)
    return output_field

def pde_filter_design_density(input_field, output_field):
    """Filter active design variables and normalize by the
        filtered mask."""
    # Active indicator M is filtered to create the denominator H(M)
    filter_work.vector()[ActiveDV] = input_field.vector()[ActiveDV]
    pde_filter_raw(filter_work, output_field)

    # rho_f = H(M rho) / H(M); passive cells restored after
    # filtering
    output_field.vector()[:] /= filter_denominator_values
    output_field.vector()[PassiveDV] = input_field.vector()[
        PassiveDV]
    return output_field

def pde_filter_design_gradient(input_field, output_field):
    """Apply the transpose of the active mask-normalized density
        filter."""
    # Adjoint of rho -> H(M rho) / H(M), restricted to active DOFs
    filter_work.vector()[ActiveDV] = input_field.vector()[ActiveDV]
    filter_work.vector()[ActiveDV] /= filter_denominator_values[ActiveDV]
    pde_filter_raw(filter_work, output_field)

    output_field.vector()[PassiveDV] = 0.0
    return output_field

```

4.7.2 Heaviside Projection

While the Helmholtz filter successfully imposes a minimum length scale, it inherently smooths the design field, resulting in a large gray transition zone between the solid and fluid regions. To recover a crisp, discrete topology, a continuous Heaviside projection is applied to the filtered variable [45, 23]:

$$\Pi_{\beta}(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad (4.16)$$

where $\eta \in [0, 1]$ is the projection threshold (typically 0.5), and β is a steepness parameter. As $\beta \rightarrow \infty$, the function approaches a strict Heaviside step function.

4.8 Algorithmic Workflow and Solver Implementation

Executing the optimization requires a highly robust discrete, iterative algorithm. To support the severe non-linearities of the turbulent regime, the solver architecture natively manages the segregated Spalart-Allmaras turbulence model, explicit physics decoupling via Picard iterations, and deeply hybridized forward flow routines.

4.8.1 Boundary Conditions and Turbulence Initialization

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

Furthermore, the solver dynamically handles pressure outlets coupled with optional velocity-component constraints, allowing the flow field to organically adapt to the evolving structural viscous dissipation without artificial boundary blockages.

4.8.2 SA Numerical Solution Controls

Although Streamline Upwind Petrov-Galerkin stabilization is a common treatment for advection-dominated Galerkin formulations [14], the computations reported in this thesis do not rely on SUPG terms or pseudo-time substepping in the SA equation. Instead, numerical control is provided through Picard-level coupling, carefully bounded turbulence variables, continuation of the topology penalties, and the hybrid forward-flow recovery strategy described below.

4.8.3 Hybrid Forward Flow Recovery and Picard Decoupling

Rather than relying solely on simple fixed-point decoupled sweeps, the forward fluid solver utilizes an advanced hybrid architecture integrating both Incremental Pressure Correction Scheme (IPCS) methods and monolithic SNES workflows.

During the optimization loop, a Picard iteration scheme strictly decouples the fluid and turbulence physics. The flow solver advances the momentum field using the eddy viscosity from the previous iteration, and this updated velocity is then passed back to the SA solver. During volatile intermediate design steps, the fluid framework initiates convection continuation strategies and leverages Stokes-Brinkman warm starts to slowly reintroduce inertial physics. If convergence stalls, the SNES solver autonomously steps through multiple recovery schedules (including dynamic tolerance reductions and altered line-search methods).

The optimization architecture supports non-converged acceptance, allowing the algorithm to passively absorb intermediate errors and maintain topological momentum. However, prior to the continuous adjoint assembly, optional configured flow

solves are executed, becoming strictly bound where enabled. When utilized, these strict solves confirm that the sensitivities, and specifically the frozen turbulent fields, are mathematically derived from a fully equilibrated state. Otherwise, relying on the accepted residual criteria serves as a necessary pragmatic compromise to maintain solver momentum in highly separated regimes.

4.8.4 Python Implementation and Continuation Strategy

To ensure convergence to a physically meaningful binary design, the algorithm iterates over a strict continuation schedule. As the optimization progresses, the Brinkman penalty parameter (q) and the projection steepness (β) are iteratively increased, while the MMA move limits are proportionally tightened.

The global optimization routine, reflecting the hybrid solver calls, explicit gradient filtering, frozen adjoint solve, and penalty schedules, is outlined in Listing 4.2 as a schematic pseudocode. Note that while the repository source code uses the variable name `rho` for the design density, it is denoted here as `rho_raw` to explicitly avoid confusion with the fluid mass density ρ .

LISTING 4.2: Schematic pseudocode representing the algorithmic architecture of the iterative topology optimization loop using the frozen turbulence assumption. Code structure has been simplified to highlight the core physics, hybrid solvers, and adjoint pipelines.

```

for stage_idx, q_val in enumerate(Q_PENAL_SCHEDULE):
    # Update continuation parameters for the current stage
    BETA_PROJ.assign(BETA_PROJ_SCHEDULE[stage_idx])
    q_penal.assign(q_val)

while inner_count < max_iters_now and not objective_converged:
    # 1. Spatial Filtering & Penalized Wall-Distance Update
    rho_f = pde_filter_design_density(rho_raw, rho_f)
    update_wall_distance_field()

    if iter_count == 0:
        initialize_forward_guess_with_stokes()

    # 2. Forward Physics (Outer Picard Loop for Decoupling)
    for picard_idx in range(PICARD_STEPS):
        solve_forward_picard(...) # Robust intermediate flow solve

    sa_model.construct_forms(w_fwd.sub(0))
    sa_model.solve_turbulence_model()
    sa_model.update_variables(relaxation=TURBULENCE_RELAXATION)

    # Lock in the updated eddy viscosity for the frozen assumption
    nu_tilde_frozen.assign(sa_model.nu_tilde0)

    # Final configured flow solve, strict where enabled
    solve_forward_snes(...)

    # 3. Sensitivity Analysis (Continuous Frozen Adjoint)
    solve_adjoint(objective_adjoint_form)

```

```
# 4. Gradient Assembly & Transpose Filtering
unfiltered_gradient.vector()[:] = assemble(objective_ddx)[:]
filtered_gradient = pde_filter_design_gradient(
    unfiltered_gradient, filtered_gradient)

# 5. Method of Moving Asymptotes (MMA) Design Update
(xmma, ...) = mmassub(mmma, num_mma, iter_count, xval, xmin,
    xmax, ...)

# Update raw density array with strict bounding
rho_raw.vector()[ActiveDV] = xmma[:, 0]

inner_count += 1
```

4.9 Conclusion

This chapter has established the mathematical and algorithmic framework used for the turbulent topology optimization results in this thesis. By employing the Fictitious Domain Method with Brinkman momentum resistance [13, 21], the evolving structural topology is integrated into the fixed computational grid. The SA working-variable and reciprocal wall-distance equations are coupled to the topology through the power-law penalty, while the DG0 Helmholtz interior penalty filter [28] and continuous Heaviside projection [45] enforce a minimum length scale and reduce mesh-dependent numerical artifacts.

The optimization is driven by the frozen-turbulence continuous adjoint [47, 35]. This assumption is defensible as a stable engineering approximation, but the thesis explicitly treats it as an approximation whose limitations must be checked through finite-difference sensitivity verification and independent CFD validation. Embedded within a Picard-decoupled Python architecture utilizing hybrid IPCS/SNES recovery schemes, this methodology transforms the verified fluid dynamics engine from Chapter 3 into the automated design tool evaluated in Chapter 5.

Chapter 5

Topology Optimization for Turbulent Flows: Results and CFD Validation

5.1 Introduction

Building upon the mathematical and numerical framework developed in Chapter 4, this chapter evaluates the turbulent topology optimization solver on established internal-flow benchmarks. The main emphasis is the Alexandersen pipe bend, because this is the working configuration used for the final thesis campaign: **reciprocal_penalized** wall distance, **design** wall-density source, power-law SA and wall-distance penalties, and the frozen-turbulence adjoint. The chapter therefore first verifies the sensitivity calculation for this exact Alexandersen setup before discussing the topology optimization result.

The evaluation addresses three questions. First, does the implemented frozen adjoint produce derivatives that agree with central finite differences for the Alexandersen pipe bend? Second, does the resulting optimized topology reproduce the expected smooth, separation-avoiding bend behavior reported in the turbulent topology optimization literature? Third, does the extracted diffuse design remain credible when re-evaluated as a sharp geometry in Ansys Fluent with SA and SST $k\text{-}\omega$ turbulence models?

The Dilgen and Yoon cases are retained as supporting benchmarks. They document how the same solver family behaves on other turbulent topology optimization examples, but they are not used as the primary sensitivity verification for the final Alexandersen pipe-bend results. To maintain focus on the core turbulent physics and topology results, detailed geometric parameters, solver settings, and Ansys grid-independence studies are deferred to Appendix B. Similarly, the detailed laminar-versus-turbulent topology comparisons are consolidated in Appendix C.

5.2 Benchmark Strategy and Evaluation Metrics

Table 5.1 outlines the strategic role of each benchmark. The Alexandersen pipe bend is the primary case: it receives the finite-difference sensitivity verification, the main topology optimization discussion, and the independent Ansys validation. The remaining cases provide supporting evidence and context.

TABLE 5.1: Benchmark strategy used in Chapter 5. Detailed physical and numerical parameters are listed in Appendix B.

Case	Role in this thesis	Reference comparison	com-	Validation scope
Alexandersen pipe bend	Primary thesis case: frozen-adjoint sensitivity verification followed by topology optimization at $V_f = 0.25$	Pipe-bend setup and optimized fields from Bayat et al. [11]		Five-cell central finite-difference check; FEniCS optimization; Ansys-SA and Ansys-SST validation
Dilgen pipe bend	Supporting fixed-geometry diagnostic retained for comparison with the earlier Dilgen-style verification workflow	Velocity contours and SA sensitivity plots from Dilgen et al. [17]		FEniCS sensitivity comparison only
Dilgen U-bend	Supporting topology optimization case for a curved turbulent manifold	Objective and volume histories; SA topology and velocity contours from Dilgen et al. [17]		FEniCS and, if time permits, Ansys validation
Yoon diffuser	Supporting adverse-pressure-gradient benchmark	Geometry, boundary conditions, and turbulent layout from Yoon [48]		FEniCS and, if time permits, Ansys validation

5.2.1 Optimization and Validation Metrics

Throughout the optimization process, the objective J follows the case-specific metric defined in Chapter 4. Dissipation-driven cases use the Brinkman-penalized power dissipation in Eq. 4.10, while the Alexandersen pipe bend uses the average inlet-pressure objective in Eq. 4.11. The dissipation formulation smoothly combines physical viscous dissipation with the artificial resistance necessary to enforce solid regions. However, this diffuse metric cannot be directly compared to the results from Ansys Fluent. Once the geometry is extracted, the Brinkman penalty is removed, leaving discrete, sharp solid walls.

To bridge this gap, the primary validation metric is the physical viscous dissipation integrated over the active fluid domain,

$$\Phi_D = \int_{\Omega_{\text{design}}} \frac{1}{2} \mu_{\text{eff}} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) : (\nabla \mathbf{u} + \nabla \mathbf{u}^T) d\Omega, \quad (5.1)$$

where the effective viscosity is $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent evaluations and $\mu_{\text{eff}} = \mu$ for laminar reference solves. Furthermore, because pressure drop is the most direct and practical engineering interpretation of these flow losses, Δp is also reported from the Ansys validation runs. All two-dimensional dissipation quantities are evaluated per unit depth.

5.2.2 Wall-Distance and SA Penalization Conventions

The final Alexandersen computations use one wall-distance and penalty convention throughout: **reciprocal_penalized** wall distance, **design** wall-density source, and the power-law solid indicator in Eq. 4.5. The wall-distance field is updated in the forward loop, but it is treated as a frozen coefficient during the adjoint sensitivity solve. Thus, the finite-difference verification below checks the implemented frozen derivative for the same practical assumption used by the optimizer.

5.3 Dilgen Pipe Bend: Supporting Sensitivity Diagnostic

The Dilgen pipe bend [17] is retained as a supporting diagnostic because it was used during development to compare frozen-adjoint sensitivities against central finite differences on a known turbulent bend. It is not the primary sensitivity verification for the final thesis campaign. The primary check is the Alexandersen pipe bend in Section 5.5.2, because that case uses the same **reciprocal_penalized**, power-law, frozen-turbulence configuration as the reported optimization.

5.3.1 Geometry and Boundary Conditions

The geometric domain and boundary conditions are configured according to the pipe-bend setup illustrated in Fig. 2 of Dilgen et al. [17]. While the original authors compared multiple turbulence models, the present implementation restricts its scope exclusively to the SA model. In Dilgen et al., the channel-bend verification is initialized with $\gamma = 1$ throughout the design domain, and the SA distance function is not penalized for that verification. The present **passive** wall-density source and **poisson** wall-distance mode reproduce that choice.

5.3.2 Sensitivity Verification Results

To ensure the baseline flow physics are captured correctly, the forward FEniCS-SA velocity-magnitude contours are first computed (Fig. 5.2). The main high-speed stream follows the curved passage from inlet to outlet, while the inner recirculating part of the bend remains a low-velocity region. This reproduces the qualitative structure of the SA result reported by Dilgen et al. [17] and provides the necessary forward-state basis for interpreting the adjoint field.

With the flow field established, the spatial sensitivity field $dJ/d\gamma$ is evaluated. The coordinate-wise finite-difference verification follows the central-difference metric

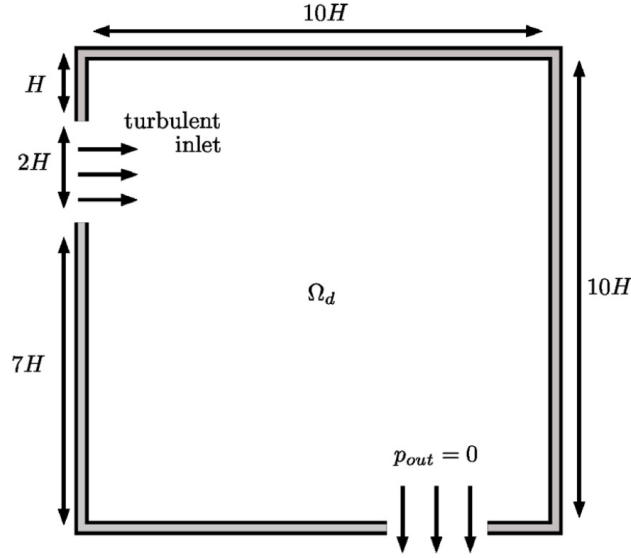


FIGURE 5.1: Geometry and boundary conditions for the Dilgen pipe-bend sensitivity verification case. The setup is reproduced from Dilgen et al. [17]; the present calculations use the SA turbulence model only.

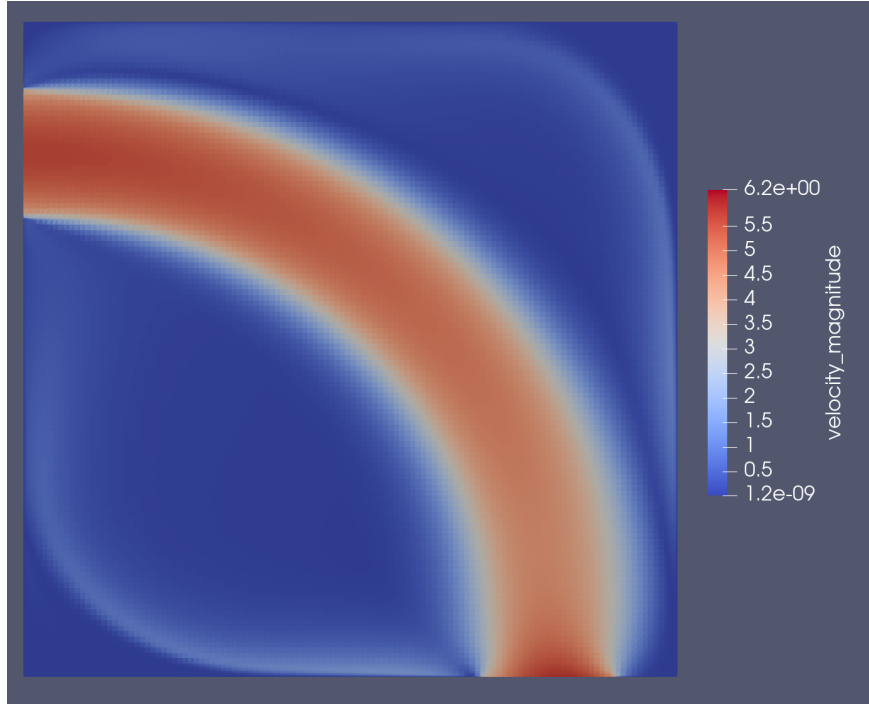


FIGURE 5.2: Velocity-magnitude contours for the Dilgen pipe bend computed with the FEniCS-SA frozen-turbulence implementation, following the contour presentation of Dilgen et al. [17].

used by Dilgen et al. [17]. For a sampled DG0 design cell i with unit perturbation vector $\mathbf{e}_i$, the frozen-turbulence finite-difference sensitivity used in the present table is

$$S_i^{\text{FD}} = \frac{J(\gamma + \Delta h \mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{v}^0) - J(\gamma - \Delta h \mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{v}^0)}{2\Delta h}, \quad \Delta h = 10^{-6}. \quad (5.2)$$

Here J is the power-dissipation objective, corresponding to Dilgen’s notation C , the superscripts $+$ and $-$ denote the converged flow states for the positively and negatively perturbed designs, and $\tilde{v}^0$ is the unperturbed SA working-variable field held fixed in the frozen-turbulence comparison. The discrepancy reported in Table 5.2 is computed as

$$\varepsilon_i^{(m)} = \frac{|S_i^{(m)} - S_i^{\text{FD}}|}{\max(|S_i^{(m)}|, |S_i^{\text{FD}}|, 10^{-30})}, \quad (5.3)$$

where m denotes the adjoint mode being compared. The present table reports only the frozen-turbulence mode. For this fixed Poisson wall-distance case, the wall distance is passive with respect to the design field; there is no penalized wall-distance derivative, no reciprocal wall-distance state, and no G -adjoint contribution.

TABLE 5.2: Coordinate sensitivity verification for the Dilgen pipe bend, following the comparison structure of Dilgen et al. Table 2 [17]. The finite-difference values are computed with Eq. (5.2); the error columns use Eq. (5.3).

Cell	Finite difference S_i^{FD}	Frozen adjoint S_i^{fr}	Absolute error	Relative error $\varepsilon_i^{\text{fr}}$
CV_1	$-1.64230238 \times 10^{-2}$	$-1.64230269 \times 10^{-2}$	3.03×10^{-9}	1.85×10^{-7}
CV_2	$-8.88178420 \times 10^{-9}$	$-4.05592879 \times 10^{-9}$	4.83×10^{-9}	5.43×10^{-1}
CV_3	$-5.09747711 \times 10^{-3}$	$-5.09745305 \times 10^{-3}$	2.41×10^{-8}	4.72×10^{-6}
CV_4	$4.22981206 \times 10^{-4}$	$4.23061639 \times 10^{-4}$	8.04×10^{-8}	1.90×10^{-4}
CV_5	$-3.05915293 \times 10^{-5}$	$-3.05958417 \times 10^{-5}$	4.31×10^{-9}	1.41×10^{-4}

The absolute agreement in Table 5.2 is strong: the largest absolute discrepancy is 8.04×10^{-8} . The only large relative error occurs at CV_2 , where both the finite-difference and adjoint sensitivities are close to zero; the absolute error there remains only 4.83×10^{-9} . Excluding this near-zero normalization case, the largest relative discrepancy is 1.90×10^{-4} . The coordinate check therefore confirms that the implemented frozen-turbulence adjoint is consistent with central finite differences for the fixed Dilgen wall-distance formulation.

Figure 5.3 shows the corresponding spatial sensitivity field. The negative sensitivity band follows the energetic curved flow path, indicating where increasing the local fluid indicator most strongly reduces the dissipation objective. By contrast, regions outside the active passage are weakly positive or close to neutral, consistent with their limited contribution to the useful flow route.

Further quantitative verification is provided by the one-dimensional cuts in Fig. 5.4. Along the diagonal line $x = y$, the minimum sensitivity is approximately

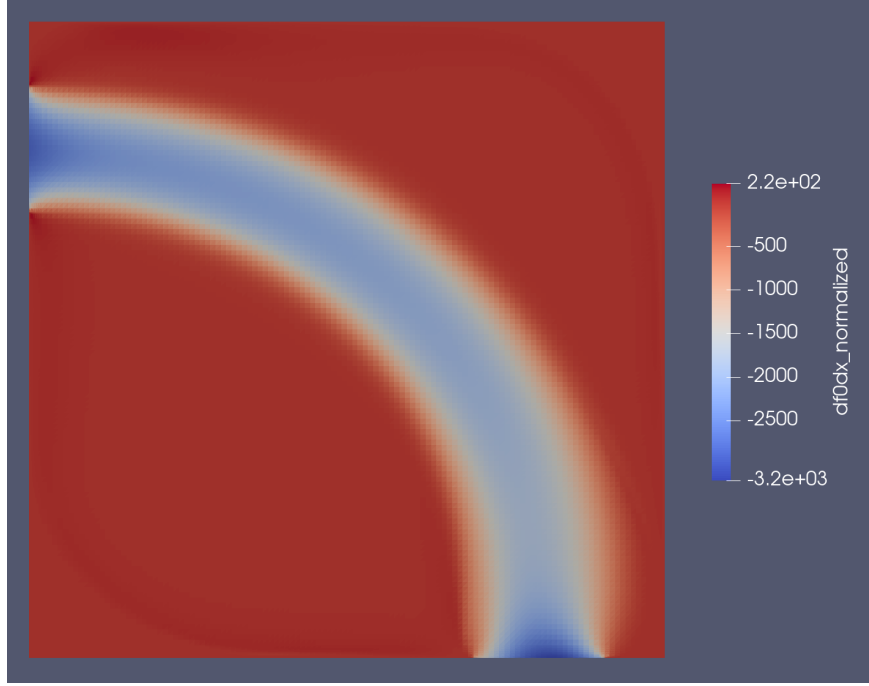


FIGURE 5.3: Frozen-turbulence sensitivity contour $dJ/d\gamma$ for the Dilgen pipe bend, following the comparison style of Dilgen et al. Fig. 4 [17].

-7.17×10^{-2} at $x = y \approx 0.575$. Along the horizontal line $y = 0.5$, the minimum is approximately -7.00×10^{-2} at $x \approx 0.65$. In both cuts, the strong negative trough appears where the sampling line intersects the accelerated bend flow, while the sensitivity approaches zero outside the active passage.

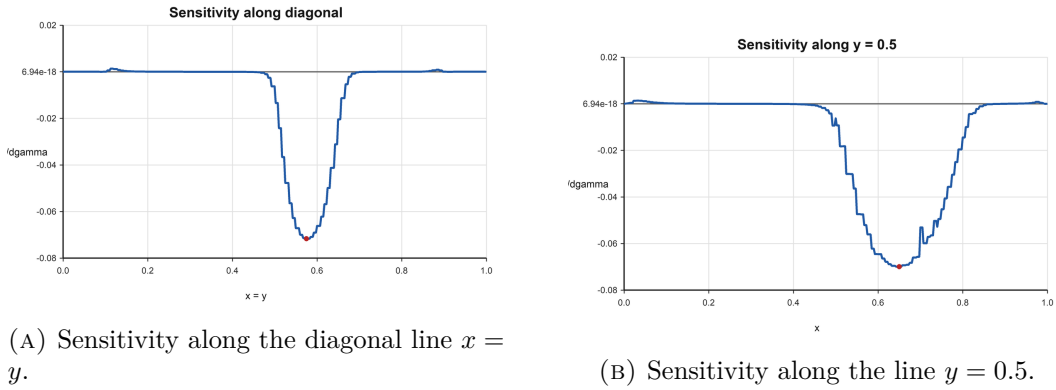


FIGURE 5.4: One-dimensional sensitivity cuts for the Dilgen pipe bend, corresponding to the quantities shown in Figs. 5 and 6 of Dilgen et al. [17].

Because this case functions purely as a fixed-domain diagnostic tool, it requires no independent Ansys validation or laminar comparison. The use of a passive wall-distance source is appropriate here because the Dilgen pipe bend is an all-fluid

sensitivity benchmark with fixed geometric walls. The conclusion from Table 5.2 is therefore limited but useful: the frozen-turbulence derivative assembled by the FEniCS implementation is consistent with central finite differences for the fixed Dilgen-style diagnostic. The final thesis verification, however, is the Alexandersen check in Section 5.5.2.

5.4 Dilgen U-Bend

The Dilgen U-bend is retained as a supporting topology optimization benchmark. By allowing the design variable to actively sculpt the flow path, the solver must dynamically update the SA wall source to reflect the newly forming turbulent boundaries.

The U-bend is not repeated as a sensitivity-verification case. It uses the same frozen-turbulence differentiation strategy as the rest of the reported optimization campaign, and the implementation check is already provided by the Alexandersen pipe bend.

5.4.1 Geometry and Boundary Conditions

The geometric domain and boundary conditions match those of Fig. 7 in Dilgen et al. [17]. The solver relies on the `poisson_penalized` wall-distance mode alongside a design-driven SA wall source.

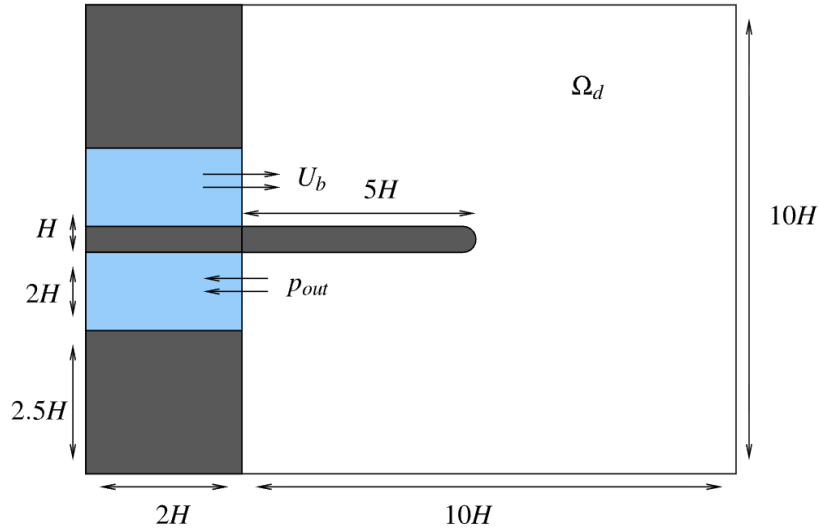


FIGURE 5.5: Geometry and boundary conditions for the Dilgen U-bend topology optimization case, reproduced from Dilgen et al. [17].

5.4.2 FEniCS Optimization Results

The convergence histories for the normalized objective function and the volume constraint are detailed in Fig. 5.6.

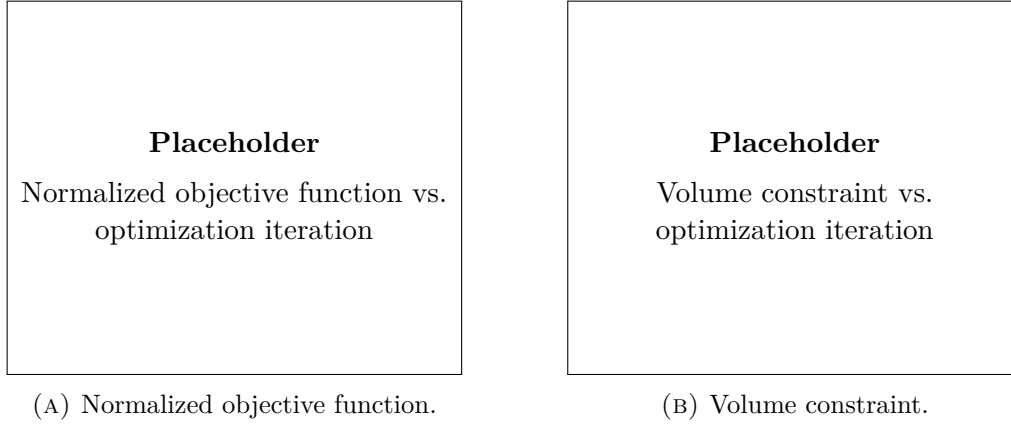


FIGURE 5.6: Optimization convergence history for the Dilgen U-bend, corresponding to the progression reported in Dilgen et al. Fig. 8 [17].

As shown in Fig. 5.7, the frozen-turbulence topology is expected to suppress the sharp inner corners that are typical of creeping-flow designs. Instead, the turbulent optimization should favor a smoothly sweeping inner radius, mitigating flow separation and reducing turbulent mixing losses. While minor discrepancies with the reference literature may arise due to the specific FEniCS continuation schedules or wall-distance implementations, the fundamental routing strategy and velocity acceleration zones are expected to align robustly with the published behaviors.

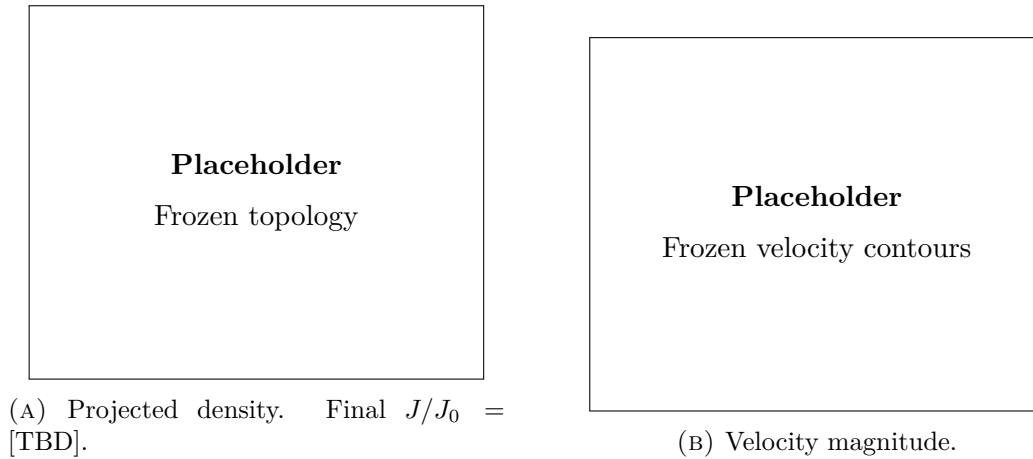


FIGURE 5.7: Final frozen-turbulence topology and velocity-magnitude contours for the Dilgen U-bend.

5.4.3 CFD Validation and Turbulence Model Sensitivity

To guarantee that the optimized diffuse geometry performs well in a realistic engineering context, the final design is extracted as a discrete, sharp-walled domain and evaluated in Ansys Fluent. As summarized in Table 5.3, the performance is analyzed using both the SA and the SST k - ω models. This dual-model approach confirms whether the design’s superior performance is a genuine physical improvement or merely an artifact of exploiting the specific formulation of the SA model during optimization.

TABLE 5.3: CFD validation metrics for the extracted Dilgen U-bend topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	[TBD]	[TBD]	Cross-code comparison
Ansys sharp geometry	SST k - ω	[TBD]	[TBD]	Turbulence-model sensitivity

5.4.4 Comparison with Laminar Design

The comparison against a purely laminar design is presented in Section C.2 of Appendix C. This analysis illustrates a central premise of this thesis: topologies optimized for the creeping-flow limit ($Re = 1$) often produce overly direct flow paths that suffer severe penalties, such as massive separation and intense turbulent dissipation, when forced to operate under turbulent flow conditions.

5.5 Alexandersen Pipe Bend

The Alexandersen pipe bend benchmark, originally formulated by Bayat, Li, and Alexandersen [11], introduces a critical cross-model verification. While the reference study employs a standard k - ε RANS model with wall functions, the present framework tackles the exact same problem using a wall-resolved SA model under the frozen-turbulence assumption. This comparison provides valuable insight into whether the fundamental topological solution, a streamlined and separation-avoiding bend, remains consistent across fundamentally different turbulence closures.

5.5.1 Geometry and Boundary Conditions

The boundary conditions and geometric constraints replicate the setup shown in Fig. 10 of Bayat et al. [11], specifically targeting the $V_f = 0.25$ volume-fraction case. In contrast to the dissipation-driven cases, the Alexandersen optimization minimizes the average inlet pressure while holding the outlet pressure fixed, so the objective directly represents the pressure loss required to drive the imposed flow through the evolving bend.

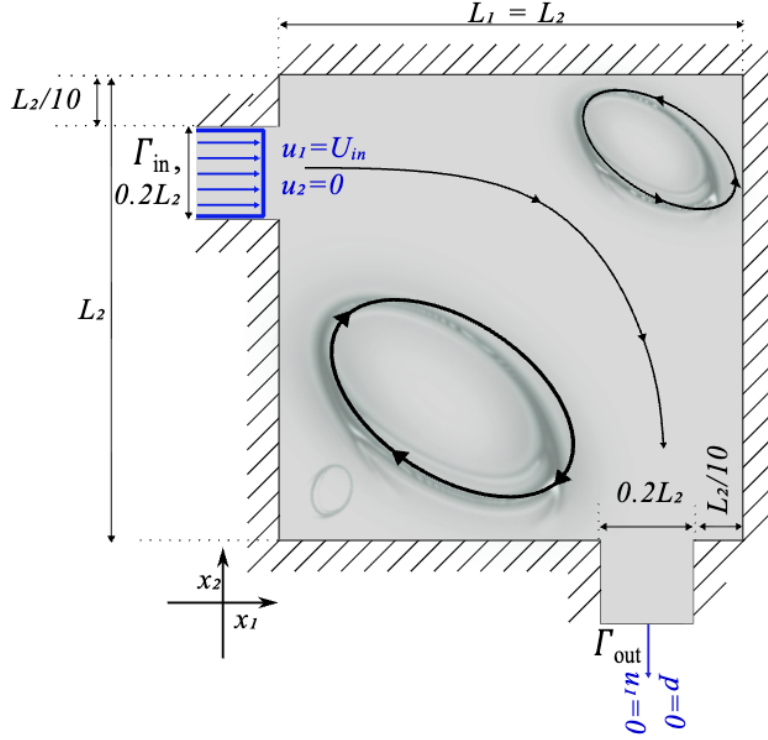


FIGURE 5.8: Geometry and boundary conditions for the Alexandersen pipe-bend benchmark, based on Bayat et al. [11]. The present evaluation targets the $V_f = 0.25$ case using the SA model.

5.5.2 Sensitivity Verification

Before interpreting the optimized topology, the frozen adjoint is verified directly on the Alexandersen pipe-bend configuration. This is the relevant check for the final thesis setup because it uses the same power-law SA penalty, `reciprocal_penalized` wall distance, `design` wall-density source, inlet-pressure objective, and frozen-turbulence adjoint used during the optimization.

Five active DG0 design cells are sampled deterministically from a random seed. For each sampled cell i , the raw density is perturbed by a central finite-difference step $\Delta h = 10^{-6}$ while the SA working-variable field and reciprocal wall-distance field are held fixed. The perturbed velocity-pressure state is then re-solved, and the finite-difference derivative is computed as

$$S_i^{\text{FD}} = \frac{J_p(\gamma + \Delta h \mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{v}^0, G^0) - J_p(\gamma - \Delta h \mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{v}^0, G^0)}{2\Delta h}. \quad (5.4)$$

This verifies the derivative actually supplied to MMA: the flow adjoint accounts for the dependence of the inlet-pressure objective on the flow solution and the density penalties, while the turbulence and wall-distance updates are frozen during differ-

entiation. The absolute and relative errors are

$$\Delta_i^{\text{abs}} = \left| S_i^{\text{fr}} - S_i^{\text{FD}} \right|, \quad \varepsilon_i^{\text{fr}} = \frac{\Delta_i^{\text{abs}}}{\max(|S_i^{\text{fr}}|, |S_i^{\text{FD}}|, 10^{-30})}. \quad (5.5)$$

TABLE 5.4: Coordinate sensitivity verification for the Alexandersen pipe bend.

Cell	Finite difference S_i^{FD}	Frozen adjoint S_i^{fr}	Absolute error Δ_i^{abs}	Relative error $\varepsilon_i^{\text{fr}}$
CV_1	[TBD]	[TBD]	[TBD]	[TBD]
CV_2	[TBD]	[TBD]	[TBD]	[TBD]
CV_3	[TBD]	[TBD]	[TBD]	[TBD]
CV_4	[TBD]	[TBD]	[TBD]	[TBD]
CV_5	[TBD]	[TBD]	[TBD]	[TBD]

After running the configured finite-difference check, the numerical values in Table 5.4 are filled from `SensitivityCheckLog.tsv` in the Alexandersen frozen-results directory.

Agreement in Table 5.4 is the primary sensitivity verification used in this chapter. The check is intentionally limited to one representative case because it verifies the implementation of the frozen derivative; repeating the same derivative check for the Alexandersen U-bend would test the same algebraic path rather than a fundamentally new sensitivity implementation.

5.5.3 FEniCS Optimization Results

Figure 5.9 illustrates the projected density field $\bar{\gamma}$ and the corresponding velocity magnitude. Despite the differing turbulence models, it is anticipated that the FEniCS-SA optimizer will converge on a broadly similar routing strategy as the $k\text{-}\varepsilon$ reference. The primary high-speed flow path should bypass the sharpest corners to minimize momentum loss, although subtle differences in the treatment of the boundary layer may result in variations in the exact gray-scale density distribution near the walls.

5.5.4 CFD Validation and Turbulence Model Sensitivity

The structural robustness of the extracted Alexandersen bend is verified in Table 5.5. Because the SST $k\text{-}\omega$ model is notably more sensitive to adverse pressure gradients than SA, evaluating the extracted geometry under both closures serves as a rigorous stress test for the optimized shape.

5.5.5 Comparison with Laminar Design

The laminar-design comparison is reported in Section C.3 of Appendix C. The expected distinction is that a laminar bend optimized at $Re = 1$ can minimize wall length and viscous shear in the creeping-flow limit, but may produce a sharper or less separation-aware turn when evaluated at the target turbulent Reynolds number.

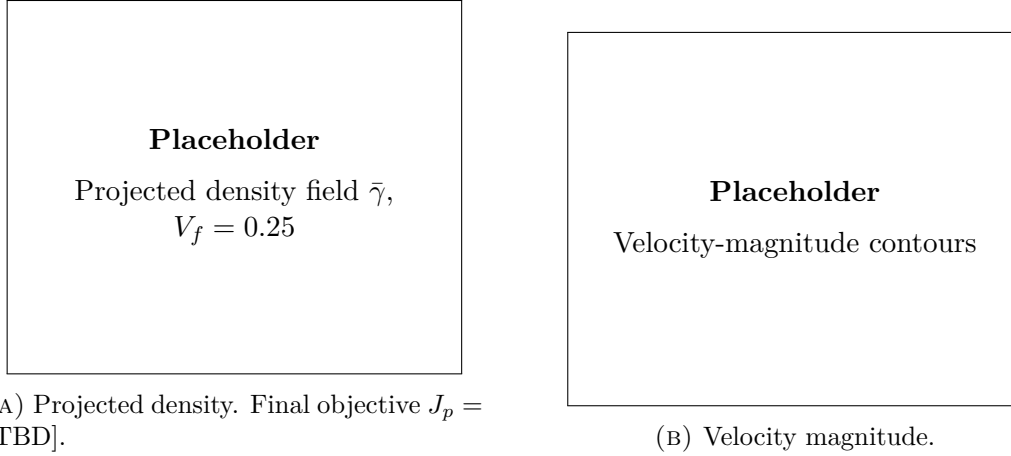


FIGURE 5.9: FEniCS-SA optimization result for the Alexandersen pipe bend at $V_f = 0.25$.

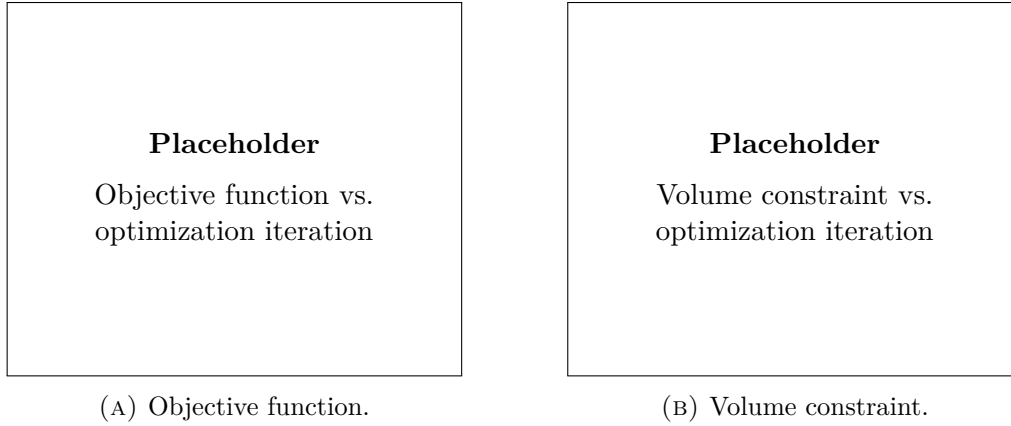


FIGURE 5.10: Optimization convergence histories for the Alexandersen pipe bend.

TABLE 5.5: CFD validation metrics for the extracted Alexandersen pipe-bend topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	[TBD]	[TBD]	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	[TBD]	[TBD]	Turbulence-model sensitivity

5.6 Yoon Diffuser

The final benchmark, based on the diffuser studied by Yoon [48], poses the most severe physical challenge in the suite. Unlike the bend-dominated flows examined previously, diffuser performance is almost entirely dictated by the management of adverse pressure gradients. This case is therefore ideal for demonstrating the optimizer’s ability to actively sculpt geometries that prevent widespread boundary layer

separation.

5.6.1 Geometry and Boundary Conditions

The configuration follows Yoon’s $Re = 3,000$ setup, characterized by $\tilde{\nu} = 1$ prescribed at the inlet and outlet boundaries. This corresponds directly to the turbulent layout reported in Fig. 22 of the original publication.

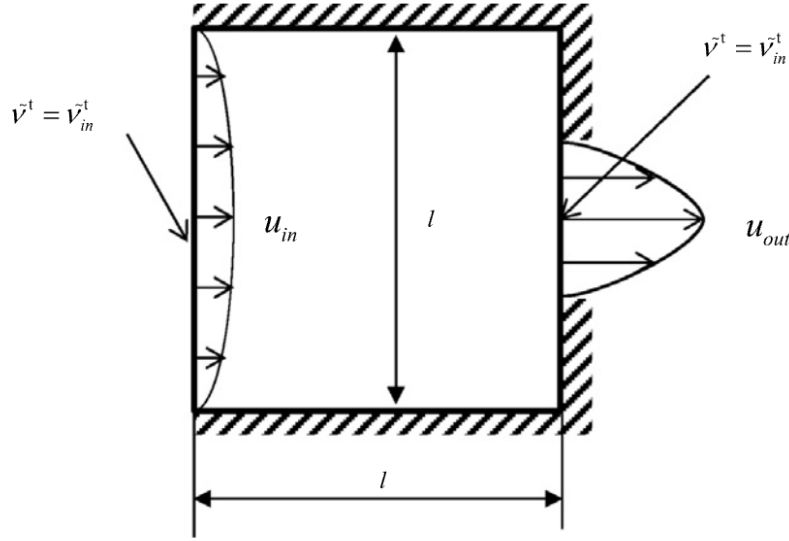


FIGURE 5.11: Geometry and boundary conditions for the Yoon diffuser benchmark, based on Yoon [48].

5.6.2 FEniCS Optimization Results

For a flow undergoing expansion, the optimization framework is expected to strongly penalize the abrupt, stepwise enlargements that often appear in Stokes-flow approximations. As seen in Fig. 5.12, the turbulent optimizer actively drives the design toward a gently diverging profile. This controlled expansion safely diffuses momentum while maintaining wall attachment, closely mirroring the characteristic trumpet-like shapes recognized in classical aerodynamics.

5.6.3 CFD Validation and Turbulence Model Sensitivity

Given that separation prediction is highly model-dependent, cross-validation in Ansys Fluent is paramount here. A diffuser design that remains attached under the SA formulation must be rigorously verified with the SST $k-\omega$ model to ensure the topology is genuinely robust against adverse pressure gradients (Table 5.6).

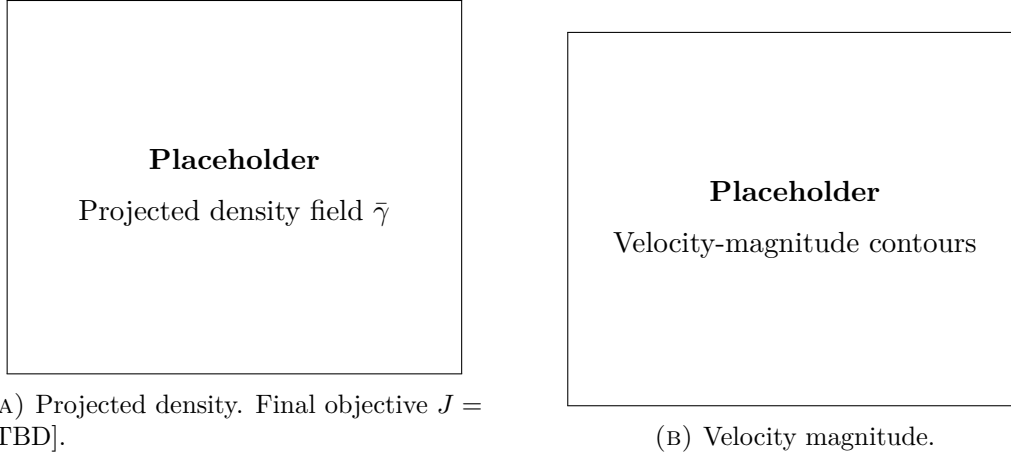


FIGURE 5.12: FEniCS-SA optimization result for the Yoon diffuser. The gracefully diverging topology prevents the massive separation zones that would otherwise dominate an abrupt expansion.

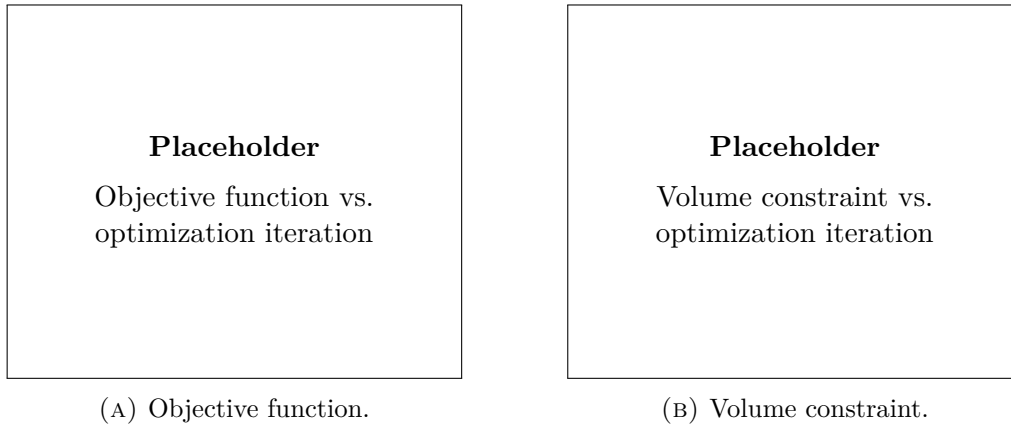


FIGURE 5.13: Optimization convergence histories for the Yoon diffuser.

TABLE 5.6: CFD validation metrics for the extracted Yoon diffuser topology.

Solver	Model	Φ_D [W/m]	Δp [Pa]	Comment
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	[TBD]	[TBD]	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	[TBD]	[TBD]	Turbulence-model sensitivity

5.6.4 Comparison with Laminar Design

As detailed in Section C.4 of Appendix C, the contrast between laminar and turbulent diffuser optimization is profound. A laminar layout minimizes viscous shear by expanding as quickly as possible, a strategy that catastrophically fails due to massive flow recirculation when subjected to high-Reynolds turbulent conditions.

5.7 Conclusion

This chapter centers the final thesis evidence on the Alexandersen pipe bend. The sensitivity verification is performed on the same `reciprocal_penalized`, power-law, frozen-turbulence configuration used for the optimization, so the finite-difference comparison directly checks the derivative supplied to MMA. The resulting topology is then evaluated as the primary turbulent design case, with the Dilgen and Yoon benchmarks retained as supporting context for curved manifolds and adverse-pressure-gradient flows.

The independent CFD validation using Ansys Fluent is therefore interpreted as a sharp-geometry check of the Alexandersen design, not merely as another internal FEniCS result. Testing the extracted topology with both SA and SST k - ω quantifies how much of the optimized performance survives a change in turbulence closure. Together with the laminar comparisons detailed in the appendices, this chapter supports the thesis argument that high-Reynolds turbulence physics must be included directly in the optimization loop when designing viable internal-flow topologies.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The primary objective of this thesis was to develop a stable, continuous turbulent topology optimization framework within FEniCS, rigorously verify its computational accuracy, and critically evaluate the physical credibility of its resulting designs. By establishing a fully independent computational fluid dynamics (CFD) re-analysis pipeline, this research moved beyond simply generating topological shapes, explicitly quantifying the fluid dynamic performance of the optimized manifolds under high-fidelity commercial turbulence closures.

Addressing the **first research question**, the theoretical limitations and computational advantages of the Spalart-Allmaras (SA) turbulence model were assessed. While its one-equation, wall-resolved formulation makes it highly tractable for continuous adjoint frameworks, its reliance on the linear Boussinesq approximation fundamentally limits its accuracy in resolving anisotropic Reynolds stresses—particularly in regimes dominated by high streamline curvature and separation.

Regarding the **second research question**, the numerical integration of this highly non-linear SA model into the FEniCS finite-element environment was successfully achieved. By deploying a rigorously decoupled Picard iteration scheme, an Incremental Pressure Correction Scheme (IPCS) for the Navier-Stokes equations, and a stabilized relaxed Eikonal equation for wall-distance computation, the forward fluid state was robustly driven to equilibrium without requiring artificial numerical diffusion.

Building on this foundation, the mathematical framework for topology optimization was constructed. The implementation confirmed that the "frozen turbulence" adjoint assumption, which treats the turbulent eddy viscosity and reciprocal wall-distance fields as fixed coefficients during gradient assembly, provides a stable and computationally efficient optimization engine. This assumption is approximate, but the Alexandersen pipe-bend finite-difference check verifies that the implemented frozen derivative is internally consistent for the configuration used in the final optimization campaign.

Finally, answering the **third research question**, the framework was subjected

to strict cross-code CFD validation. The Alexandersen pipe bend served as the primary final case: its diffuse FEniCS-SA topology was extracted as a sharp geometry and re-analyzed in Ansys Fluent. The comparative analysis in Appendix C further supports the engineering motivation for turbulent optimization by showing why laminar-optimized layouts can perform poorly at high Reynolds numbers.

Re-evaluating the extracted geometry using the Ansys SST $k - \omega$ model quantified the sensitivity of the Alexandersen design to the underlying turbulence physics. This confirmed that topologies optimized under isotropic, one-equation assumptions must be checked carefully when subjected to secondary flows, curvature effects, and anisotropic shear layers captured more strongly by two-equation models.

6.2 Limitations

An objective evaluation of this methodology reveals several important limitations that must be acknowledged:

1. **The Frozen Adjoint Approximation:** By ignoring the derivative of the eddy viscosity with respect to the design field, the optimizer cannot mathematically anticipate future turbulence production. In flows dominated by massive separation, this can blind the optimizer to the downstream hydrodynamic consequences of upstream geometric changes, potentially leading to sub-optimal local minima.
2. **Geometry Extraction Uncertainties:** The transition from a continuous, Brinkman-penalized density field ($\gamma \in [0, 1]$) to a Boolean CAD geometry introduces inherent ambiguity. The specific iso-value threshold chosen to extract the fluid volume dictates the final wall placement. Consequently, the performance discrepancies observed between FEniCS and Ansys are inextricably linked to this manual geometry extraction process, making it difficult to isolate pure solver physics from thresholding effects.
3. **Two-Dimensional Simplification:** The benchmark geometries evaluated in this thesis were restricted to 2D planar domains. Because turbulent phenomena—particularly curvature-induced secondary flows like Dean vortices and turbulent energy cascades—are inherently three-dimensional, the 2D assumption artificially suppresses complex mixing mechanisms that would inevitably occur in physical prototypes.
4. **Lack of Experimental Validation:** While independent cross-code verification against commercial finite-volume solvers strongly corroborates the numerical implementation, it is not a substitute for physical validation. The true operational performance of these algorithmically generated turbulent manifolds requires future validation against experimental pressure-drop and flow-field measurements.

6.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a mathematically stable foundation for several advanced avenues of future research:

6.3.1 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [2]) or simplified flow proxies such as Darcy models [50]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [10].

However, the necessity of capturing true turbulent convection in high-performance heat sinks has driven recent advancements in the literature. Emerging efforts have introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [30], computationally reduced multi-layer thermofluid approximations [49], and standard density-based frameworks coupled to two-equation $k - \epsilon$ closures [39].

With the turbulent momentum and SA transport equations now stably implemented and verified within the current FEniCS architecture, this framework is perfectly positioned to contribute to this cutting-edge research front. Extending the current methodology to CHT will require implementing a turbulent Prandtl number (Pr_t) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers—currently dominated by laminar assumptions [25]—could be significantly enhanced by adapting this turbulent solver.

6.3.2 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable manifolds, the current 2D FEniCS implementation must be extended to three dimensions. While the governing equations and adjoint derivations remain mathematically identical, the computational cost of resolving the boundary layers ($y^+ \leq 1$) across a 3D topology optimization grid will require significant advancements in parallel processing and highly efficient linear solvers (e.g., Algebraic Multigrid preconditioners).

6.3.3 Adjoint Model Hierarchies

This thesis deliberately uses the frozen-turbulence adjoint as the robust baseline. Future frameworks should investigate stabilized ways of reintroducing selected turbulence-

production sensitivities, especially in separated flows where the Constant Eddy Viscosity assumption is least accurate. Higher-fidelity two-equation models, such as $k - \omega$, should also be investigated directly within the continuous adjoint topology optimization loop to mitigate the curvature limitations identified in Chapter 5.

Appendices

Appendix A

Configuration and Grid Independence of the Spalart-Allmaras Verification Cases

To ensure complete scientific reproducibility and to prevent the disruption of the narrative flow in Chapter 3, all comprehensive configuration matrices, domain truncation proofs, and spatial grid independence studies for the fluid benchmarks are documented in this appendix.

A.1 Verification Benchmark 1: Channel Flow

A.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height (H), and the Reynolds number (Re_H) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The complete physical setup is detailed in Table A.1.

A.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The exact numerical solver configurations, convergence tolerances, and relaxation factors utilized in FEniCS are summarized in Table A.2.

A.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-

A. CONFIGURATION AND GRID INDEPENDENCE OF THE SPALART-ALLMARAS VERIFICATION CASES

TABLE A.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

Parameter	Channel flow setting
Benchmark type	Pressure-driven, streamwise-periodic 2D channel
Domain dimensions	Length $L = 2.0$ m; full channel height $H = 2.0$ m
Characteristic length	Full channel height $H = 2.0$ m
Reference velocity	$U_{\text{ref}} = 18.5$ m/s, utilized to define the target Reynolds number
Kinematic viscosity	$\nu = 1.81818 \times 10^{-3}$ m ² /s
Reynolds number	$Re_H \approx 2.03 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s ²
Driving condition	Fixed pressure drop from $p = 2$ Pa to $p = 0$ Pa across the periodic streamwise direction
Velocity boundary conditions	Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls
Pressure boundary conditions	$p = 2$ Pa at the upstream pressure plane; $p = 0$ Pa at the downstream pressure plane
SA boundary conditions	Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls
Initial fields	$\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 2$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m ² /s
Verification mesh	Fine_FEniCS ; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m

layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table A.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure A.1. The **Fine_FEniCS** grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. The properties of these meshes are documented in Table A.4. The wall-normal velocity profiles extracted from the Ansys solver, illustrated in Figure A.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

TABLE A.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

Parameter	Symbol	Setting
Velocity space	$\mathbf{u}$	Vector CG2, periodic in streamwise direction
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{v}$	Scalar CG1, periodic in streamwise direction
Quadrature degree	–	2
Maximum outer Picard steps	$N_{\text{Picard,max}}$	200
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-6}
Outer pressure tolerance	ϵ_p	1.0×10^{-6}
Outer SA tolerance	$\epsilon_{\tilde{v}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	5.0×10^{-3} s
Maximum IPCS steps per Picard step	–	500
IPCS velocity tolerance	–	1.0×10^{-6}
IPCS pressure tolerance	–	1.0×10^{-6}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.3
SA relaxation factor	$\alpha_{\tilde{v}}$	0.3
SA lower bound	$\tilde{v}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8} ; absolute tolerance 1.0×10^{-10} ; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: direct MUMPS; SA transport: default solver

TABLE A.3: Wall-resolved channel mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the streamwise resolution, wall-normal resolution, and inflation-layer resolution are refined.

Parameter	Coarse_FEniCS	Medium_FEniCS	Fine_FEniCS
Streamwise cells	10	20	40
Wall-normal cells	42	51	68
First-layer height Δy_1 [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
Minimum wall-normal spacing [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
Maximum wall-normal spacing [m]	7.707×10^{-2}	8.454×10^{-2}	7.589×10^{-2}
Inflation layers per wall	10	14	18
Growth ratio	1.45	1.35	1.25
Vertices	473	1,092	2,829
Triangular elements	840	2,040	5,440

A. CONFIGURATION AND GRID INDEPENDENCE OF THE SPALART-ALLMARAS VERIFICATION CASES

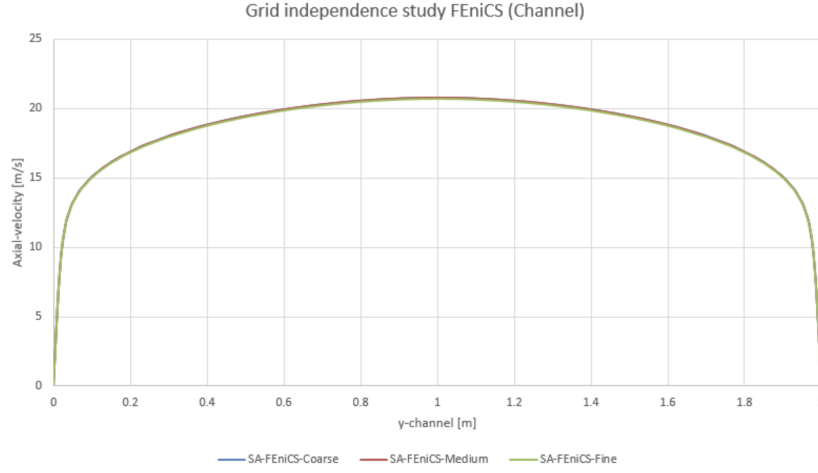


FIGURE A.1: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the periodic channel flow benchmark. The overlapping profiles demonstrate asymptotic spatial convergence.

TABLE A.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse_Ansys	Medium_Ansys	Fine_Ansys
Streamwise cells	—	—	—
Wall-normal cells	—	—	—
First-layer height Δy_1 [m]	—	—	—
Minimum wall-normal spacing [m]	—	—	—
Maximum wall-normal spacing [m]	—	—	—
Inflation layers per wall	—	—	—
Growth ratio	—	—	—
Vertices	—	—	—
Elements	—	—	—

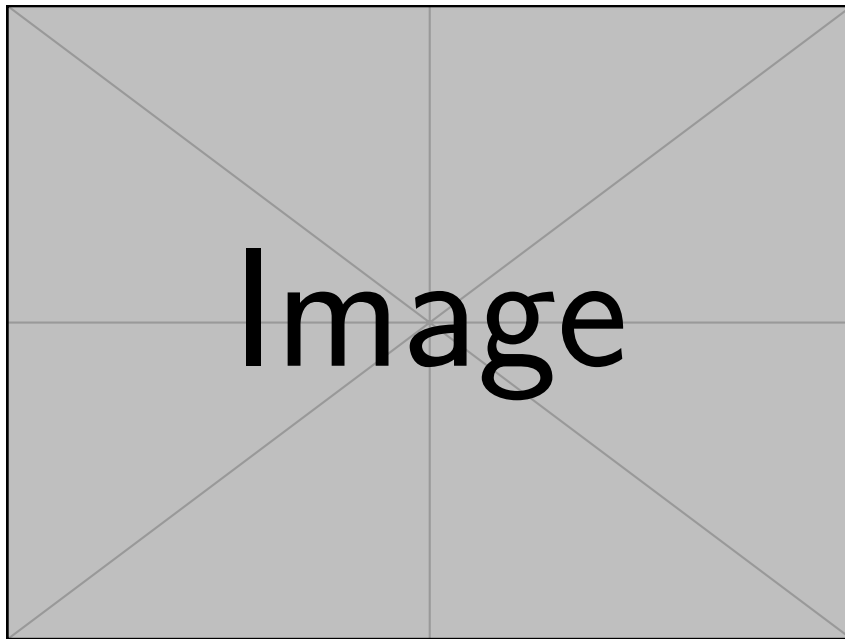


FIGURE A.2: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the Ansys periodic channel flow benchmark, demonstrating spatial convergence of the baseline data.

A.2 Verification Benchmark 2: U-Bend Geometry

A.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter (D) is explicitly retained as the characteristic length scale. This ensures the Reynolds (Re_D) and Dean (De) numbers remain non-dimensionally consistent with the commercial baseline. The configuration is detailed in Table A.5.

TABLE A.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	U-bend setting
Benchmark type	Truncated 2D symmetry-plane U-bend based on Ansys VMFL048
Domain dimensions	Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m
Characteristic length	Physical pipe diameter $D = 0.028$ m
Reference / inlet velocity	Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s
Kinematic viscosity	$\nu = 8.9 \times 10^{-7}$ m ² /s
Reynolds number	$Re_D \approx 4.47 \times 10^4$
Dean number	$De \approx 1.50 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s ²
Driving condition	Prescribed inlet velocity and zero relative static pressure at the outlet
Velocity boundary conditions	No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet
Pressure boundary conditions	$p = 0$ Pa at outlet; pressure left free at inlet and walls
SA boundary conditions	$\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m ² /s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet
Initial fields	$\mathbf{u}_0 = (0, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$
Verification mesh	Medium_FEniCS; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m

A.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table A.6.

A.2.3 Domain Truncation Study

The original 3D Ansys VMFL048 benchmark utilizes highly extended straight legs measuring 1.555 m to ensure fully developed flow. To reduce computational over-

TABLE A.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

Parameter	Symbol	Setting
Velocity space	$\mathbf{u}$	Vector CG2
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{v}$	Scalar CG1
Quadrature degree	–	4
Maximum outer Picard steps	$N_{\text{Picard,max}}$	180
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-4}
Outer pressure tolerance	ϵ_p	1.0×10^{-4}
Outer SA tolerance	$\epsilon_{\tilde{v}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	1.0×10^{-4} s
Maximum IPCS steps per Picard step	–	180
IPCS velocity tolerance	–	1.0×10^{-4}
IPCS pressure tolerance	–	2.0×10^{-3}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.1
SA relaxation factor	$\alpha_{\tilde{v}}$	0.01
SA lower bound	$\tilde{v}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8} ; absolute tolerance 1.0×10^{-10} ; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver

head for subsequent optimization stages, these legs were truncated to $30D$ (0.840 m). A preliminary sensitivity study was conducted in Ansys Fluent, comparing the full-length and truncated geometries. To mathematically prove that this reduction does not artificially alter the flow physics, the cross-sectional velocity profiles at the critical bend exit were extracted for both domains, as plotted in Figure A.3. The complete overlap of the profiles confirms that the $30D$ truncation successfully isolated the curvature physics without artificially impacting the boundary layer separation.

A.2.4 Spatial Verification and Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. A spatial grid independence study was conducted across three refined unstructured grids. The node counts, element densities, and inflation layer parameters for these generated meshes are explicitly documented in Table A.7. Similar to the channel benchmark, the wall-

A. CONFIGURATION AND GRID INDEPENDENCE OF THE SPALART-ALLMARAS VERIFICATION CASES

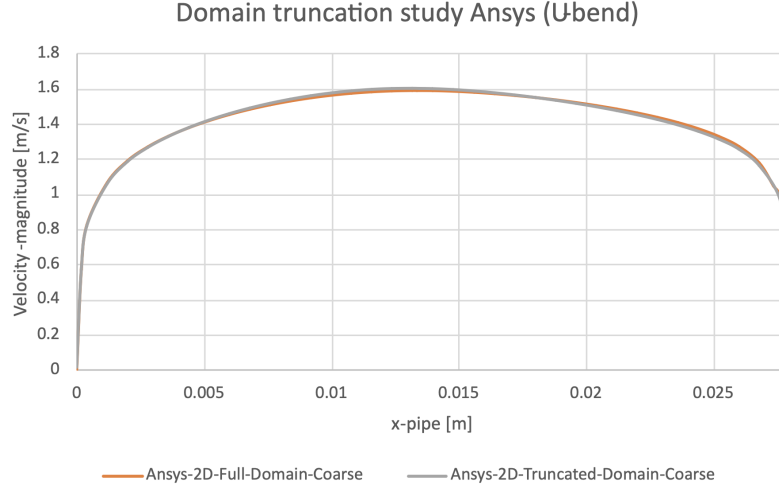


FIGURE A.3: Velocity profile comparison at the U-bend exit between the full Ansys domain (1.555 m legs) and the truncated computational domain (30D legs). The complete overlap confirms the validity of the truncated geometry.

adjacent element height was strictly maintained at 1.20×10^{-5} m to satisfy the low-Reynolds SA wall constraints, while the internal element sizing was systematically refined.

TABLE A.7: Wall-resolved U-bend mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the bulk element size and inflation-layer resolution are refined.

Parameter	Coarse_FEniCS	Medium_FEniCS	Fine_FEniCS
Bulk element size [m]	2.80×10^{-3}	1.40×10^{-3}	8.00×10^{-4}
First-layer height Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	14	18
Growth ratio	1.45	1.30	1.25
Inflation thickness [m]	1.07×10^{-3}	1.53×10^{-3}	2.62×10^{-3}
Vertices	23,751	73,978	182,450
Triangular elements	45,998	144,952	359,646

Convergence was visually verified by extracting the velocity profiles directly at the bend exit, a highly sensitive region for curvature-induced separation, as illustrated in Figure A.4. The chosen **Medium_FEniCS** grid perfectly captured the velocity gradients near the wall and provided sufficient isotropic resolution in the bulk flow to resolve the separation bubbles without numerical instability.

Similarly, to guarantee the integrity of the cross-code validation in this highly convective regime, an equivalent mesh independence study was executed for the Ansys Fluent baseline. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table A.8), the Ansys velocity

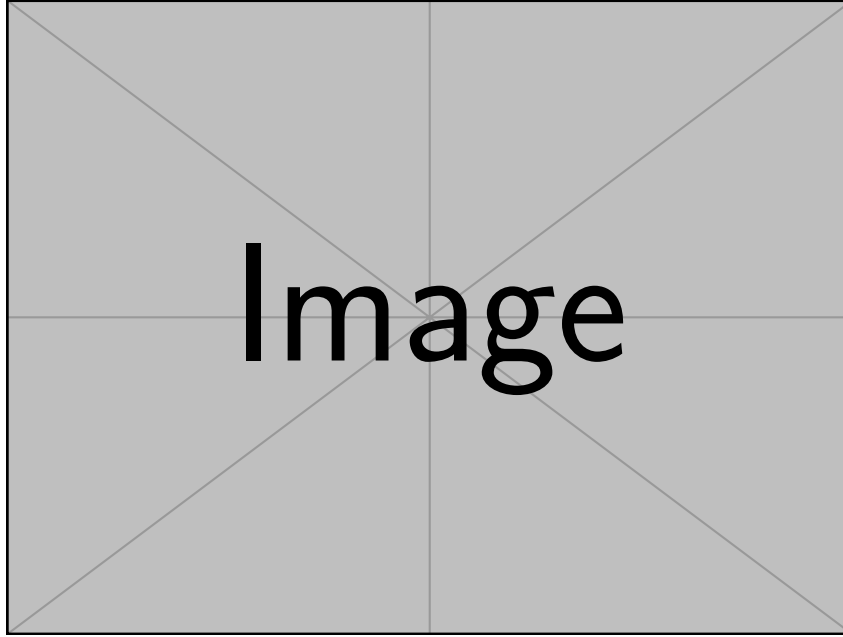


FIGURE A.4: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine grids in FEniCS. The profiles demonstrate spatial convergence, justifying the selection of the Medium grid for the final verification.

profiles at the bend exit, shown in Figure A.5, were verified to be spatially converged. This ensures that the commercial baseline reference data is independent of grid resolution.

TABLE A.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse_Ansys	Medium_Ansys	Fine_Ansys
Bulk element size [m]	–	–	–
First-layer height Δy_1 [m]	–	–	–
Inflation layers	–	–	–
Growth ratio	–	–	–
Inflation thickness [m]	–	–	–
Vertices	–	–	–
Elements	–	–	–

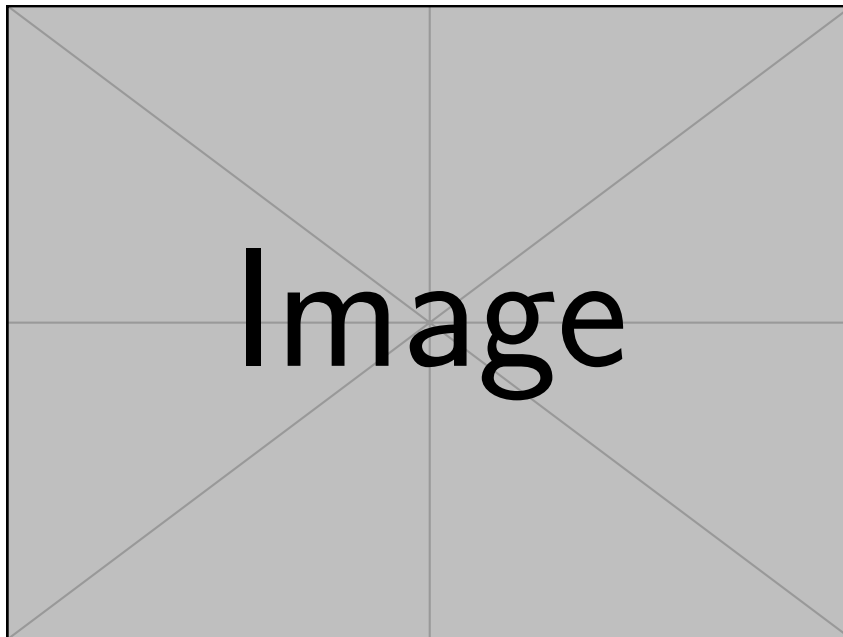


FIGURE A.5: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine Ansys grids, demonstrating spatial convergence of the commercial baseline.

Appendix B

Configuration and Grid Independence of the Topology Optimization Benchmarks

This appendix collects the detailed setup information that would interrupt the results narrative in Chapter 5. It includes the geometry, boundary-condition, physical, numerical, and validation-mesh settings for the benchmark cases. The Dilgen pipe bend is included as a sensitivity verification case. The Dilgen U-bend, Alexandersen pipe bend, and Yoon diffuser are the three true topology optimization cases and therefore include Ansys grid-independence tables for the extracted geometries.

B.1 Validation and Meshing Strategy

For the three extracted topology optimization designs, the Ansys validation meshes are refined through coarse, medium, and fine resolutions. The reported validation metrics in Chapter 5 should be taken from a mesh for which both viscous dissipation and pressure drop have become insensitive to further refinement. The final thesis should report the selected mesh explicitly for each case.

Both SA and SST $k-\omega$ validation runs should use the same extracted geometry, inlet condition, outlet condition, and fluid properties. The mesh-independence comparison should be performed for each turbulence model because the near-wall and separated-flow predictions can respond differently to grid refinement.

B.2 Wall-Distance and Penalization Conventions

The benchmark set uses more than one wall-distance and topology-penalty convention. Table B.1 records these implementation choices so that the results can be interpreted without assuming a single global wall-distance equation or penalty interpolation.

TABLE B.1: Wall-distance and SA wall-source conventions used by the Chapter 5 benchmarks.

Case	Implementation note
Dilgen pipe bend	Uses a fixed Poisson-style wall-distance field and Brinkman/RAMP-scaled interpolation for the SA working-variable penalty.
Dilgen U-bend	Uses the <code>poisson_penalized</code> wall-distance mode with a design-driven SA wall source and Brinkman/RAMP-scaled interpolation for the SA and wall-distance penalties.
Alexandersen pipe bend	Uses the <code>reciprocal_penalized</code> wall-distance mode with a design-driven SA wall source and power-law interpolation for the SA and wall-distance penalties.
Yoon diffuser	Uses the <code>reciprocal_penalized</code> wall-distance mode following the reciprocal/Yoon-style wall-distance logic of Yoon [48], with power-law interpolation for the SA and wall-distance penalties.

Chapter 4 defines these alternatives explicitly. In particular, the Dilgen-style cases should not be described as if they used the same reciprocal wall-distance equation or the same topology-penalty interpolation as the Yoon-style cases.

B.3 Benchmark 1: Dilgen Pipe Bend

The Dilgen pipe bend is used for sensitivity verification only. It does not require Ansys geometry extraction or a laminar topology comparison.

TABLE B.2: Geometric, physical, and boundary-condition parameters for the Dilgen pipe-bend sensitivity case.

Parameter	Setting
Reference case	Pipe-bend sensitivity benchmark from Dilgen et al. [17]
Role in thesis	Fixed-geometry SA sensitivity verification
Reference figure for setup	Dilgen et al. Fig. 2 (left)
Reference flow figure	Dilgen et al. Fig. 2 velocity-magnitude contours
Turbulence model in this thesis	Spalart-Allmaras only
Models compared in reference paper	SA and SST $k-\omega$
Reynolds number	[TBD]
Inlet condition	[TBD: velocity profile and SA inlet value]
Outlet condition	[TBD]
Wall conditions	No-slip velocity; SA wall condition [TBD]
Wall-distance mode	Fixed Poisson-style wall-distance field
SA penalty interpolation	Brinkman/RAMP-scaled indicator
Performance/sensitivity quantity	$dJ/d\gamma$ and the verification metrics corresponding to Dilgen et al. Figs. 5 and 6

TABLE B.3: Numerical FEniCS settings for the Dilgen pipe-bend sensitivity case.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	[TBD]
SA working-variable space	$\tilde{\nu}$	[TBD]
Design/sensitivity space	γ	[TBD]
Reference mesh	–	[TBD]
Forward coupling	–	[TBD]
Frozen adjoint state	–	$(\mathbf{u}^*, p^*)$
Coupled SA adjoint state	–	$(\mathbf{u}^*, p^*, \tilde{\nu}^*)$
Wall-distance adjoint	–	Not included for this Dilgen-style comparison
Linear solver	–	[TBD]
Convergence tolerances	–	[TBD]

B.4 Benchmark 2: Dilgen U-Bend

The Dilgen U-bend is the first true topology optimization case. It uses the same Dilgen-style wall-distance mode as the pipe-bend sensitivity case, but now the design field is active and must define the SA wall source.

TABLE B.4: Geometric, physical, and boundary-condition parameters for the Dilgen U-bend topology optimization case.

Parameter	Setting
Reference case	U-bend benchmark from Dilgen et al. [17]
Reference figure for setup	Dilgen et al. Fig. 7
Design-domain dimensions	[TBD]
Non-design fluid regions	[TBD]
Fixed solid regions	[TBD, if applicable]
Volume fraction	[TBD]
Reynolds number	[TBD]
Density and viscosity	[TBD]
Inlet condition	[TBD: velocity profile and SA inlet value]
Outlet condition	[TBD]
Wall conditions	No-slip velocity; SA wall condition [TBD]
Wall-distance mode	poisson_penalized
SA wall-density source	design
SA/wall penalty interpolation	Brinkman/RAMP-scaled indicator
Primary FEniCS plots	Objective history, volume history, topology, velocity magnitude
Validation metrics	Φ_D , pressure drop, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST

B. CONFIGURATION AND GRID INDEPENDENCE OF THE TOPOLOGY OPTIMIZATION BENCHMARKS

TABLE B.5: Numerical FEniCS solver and topology-optimization parameters for the Dilgen U-bend.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	[TBD]
SA working-variable space	$\tilde{\nu}$	[TBD]
Design-density space	γ	[TBD]
Reference FEniCS mesh	–	[TBD]
Adjoint variants evaluated	–	Frozen turbulence and coupled SA adjoint
RAMP continuation	q	[TBD]
Projection continuation	β	[TBD]
MMA move limits	–	[TBD]
Maximum optimization iterations	–	[TBD]
Filter radius	r_f	[TBD]
Forward flow solver	–	[TBD]
Frozen NS-SA coupling	–	[TBD]
Linear solver	–	[TBD]

TABLE B.6: Ansys Fluent mesh-independence table for the extracted Dilgen U-bend geometry.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
First-layer height Δy_1 [m]	[TBD]	[TBD]	[TBD]
Inflation layers	[TBD]	[TBD]	[TBD]
Growth rate	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+ , SA	[TBD]	[TBD]	[TBD]
Maximum y^+ , SST $k-\omega$	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{\text{SST } k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected validation mesh		[TBD]	

B.5 Benchmark 3: Alexandersen Pipe Bend

This case follows the pipe-bend setup of Bayat et al. [11]. The reference paper uses a standard k - ε model with wall functions; the present thesis uses the SA model and evaluates the $V_f = 0.25$ design.

TABLE B.7: Geometric, physical, and boundary-condition parameters for the Alexandersen pipe-bend topology optimization case.

Parameter	Setting
Reference case	Pipe-bend benchmark from Bayat et al. [11]
Reference figure for setup	Bayat et al. Fig. 10
Reference velocity figure	Bayat et al. Fig. 15
Reference turbulence model	Standard k - ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Volume fraction	$V_f = 0.25$
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Non-design fluid regions	Inlet and outlet lead sections of length $0.2L$ and port height/width $0.2L$; inlet over $y/L \in [0.7, 0.9]$ and outlet over $x/L \in [0.7, 0.9]$
Reynolds number	$Re = 5,000$, based on $U_{in}H_{in}\rho/\mu$ with $H_{in} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from $I = 7.5\%$ and $\ell/H_{in} = 0.05$, giving $\tilde{\nu}_{in} \approx 5.67 \times 10^{-4}$
Outlet condition	Pressure outlet $p = 0$ on the lower outlet, with $u_x = 0$ imposed on the outlet boundary
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls; topology-created solids enter the SA and wall-distance equations through the design-density penalties
Wall-distance mode	reciprocal_penalized
SA wall-density source	design , with solid threshold 0.50
SA/wall penalty interpolation	Power-law indicator with penalty amplitude 10^3 and exponent $n = 4$
Validation metrics	Average inlet pressure, pressure drop, Φ_D , and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST

B. CONFIGURATION AND GRID INDEPENDENCE OF THE TOPOLOGY OPTIMIZATION BENCHMARKS

TABLE B.8: Numerical FEniCS solver and topology-optimization parameters for the Alexandersen pipe bend.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	<code>mesh_yplus1.xdmf</code> ; 39,128 vertices, 77,334 triangles, $h_{\max} = 0.007L$, first near-wall strip sized for approximately $y^+ = 1$
Objective	J	Average inlet pressure with fixed outlet pressure $p = 0$
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and the reciprocal wall-distance field are frozen during the flow adjoint solve
RAMP continuation	q	Code schedule $q = 1/q_\alpha = [1/150, 1/75, 1/30, 1/15]$ with $\alpha_{\text{solid}} = 100$
Projection continuation	β	$[4, 6, 9, 13]$ with threshold $\eta = 0.50$
MMA move limits	–	$[0.05, 0.04, 0.03, 0.02]$
Maximum optimization iterations	–	Four continuation stages with 25 MMA iterations each, 100 maximum total; stop if objective change is below 10^{-5} for five iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.20
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves

TABLE B.9: Ansys Fluent mesh-independence table for the extracted Alexandersen pipe-bend geometry.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
First-layer height Δy_1 [m]	[TBD]	[TBD]	[TBD]
Inflation layers	[TBD]	[TBD]	[TBD]
Growth rate	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+ , SA	[TBD]	[TBD]	[TBD]
Maximum y^+ , SST $k-\omega$	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{\text{SST } k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected validation mesh		[TBD]	

B.6 Benchmark 4: Yoon Diffuser

The Yoon diffuser is the SA-based diffuser benchmark used to compare against the turbulent layout reported by Yoon [48]. The target turbulent case is $Re = 3,000$ with $\tilde{\nu} = 1$ prescribed at both inlet and outlet.

TABLE B.10: Geometric, physical, and boundary-condition parameters for the Yoon diffuser topology optimization case.

Parameter	Setting
Reference case	Diffuser benchmark from Yoon [48]
Reference figure for setup	Yoon Fig. 20
Reference turbulent topology	Yoon Fig. 22
Reynolds number	$Re = 3,000$
SA inlet/outlet value	$\tilde{\nu} = 1$ at inlet and outlet
Design-domain dimensions	[TBD]
Non-design fluid regions	[TBD]
Volume fraction	[TBD]
Density and viscosity	[TBD]
Inlet condition	[TBD]
Outlet condition	[TBD]
Wall conditions	[TBD]
Wall-distance mode	reciprocal_penalized
SA wall-density source	design
SA/wall penalty interpolation	Power-law indicator
Validation metrics	Φ_D , pressure drop, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST

B. CONFIGURATION AND GRID INDEPENDENCE OF THE TOPOLOGY OPTIMIZATION BENCHMARKS

TABLE B.11: Numerical FEniCS solver and topology-optimization parameters for the Yoon diffuser.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	[TBD]
SA working-variable space	$\tilde{\nu}$	[TBD]
Design-density space	γ	[TBD]
Reference FEniCS mesh	–	[TBD]
Adjoint treatment	–	[TBD]
RAMP continuation	q	[TBD]
Projection continuation	β	[TBD]
MMA move limits	–	[TBD]
Maximum optimization iterations	–	[TBD]
Filter radius	r_f	[TBD]
Forward flow solver	–	[TBD]
Frozen NS-SA coupling	–	[TBD]
Linear solver	–	[TBD]

TABLE B.12: Ansys Fluent mesh-independence table for the extracted Yoon diffuser geometry.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
First-layer height Δy_1 [m]	[TBD]	[TBD]	[TBD]
Inflation layers	[TBD]	[TBD]	[TBD]
Growth rate	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+ , SA	[TBD]	[TBD]	[TBD]
Maximum y^+ , SST $k-\omega$	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{\text{SST } k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected validation mesh		[TBD]	

Appendix C

Performance Comparison of Laminar and Turbulent Topologies

This appendix supports the main results in Chapter 5 by comparing topologies optimized with a laminar formulation against topologies optimized with the turbulent SA formulation. The Dilgen pipe bend is not included because it is a fixed-geometry sensitivity verification case rather than a topology optimization case. The comparison is therefore performed for the Dilgen U-bend, the Alexandersen pipe bend, and the Yoon diffuser.

C.1 Methodology and Evaluation Pipeline

Each true topology optimization benchmark is solved twice. The laminar reference design is optimized at $Re = 1$, where the optimizer primarily responds to creeping-flow viscous losses. The turbulent design is optimized at the target high-Reynolds operating condition using the SA topology optimization framework described in Chapter 4. The same design-domain restrictions, volume constraints, and inlet/outlet locations are retained wherever possible so that the topology comparison isolates the effect of the flow physics rather than a change in problem definition.

After optimization, both projected density fields are thresholded and extracted as sharp geometries. The extracted laminar and turbulent designs are then evaluated in Ansys Fluent under the same turbulent operating condition. The key performance metrics are viscous dissipation, pressure drop, and any case-specific loss measure reported in Chapter 5. The purpose of this appendix is not merely to show that the topologies look different, but to test whether the turbulent topology actually performs better when both designs are exposed to the same turbulent flow.

For fairness, the extracted laminar geometries receive their own grid-independence checks. The turbulent-design grid studies are reported in Appendix B; this appendix adds the corresponding laminar-design mesh tables and the final laminar-vs-turbulent performance comparison.

C.2 Dilgen U-Bend

The Dilgen U-bend is expected to show a clear difference between laminar and turbulent design logic. A laminar optimizer can favor short, smooth routes that reduce creeping-flow shear, while the turbulent optimizer must also account for momentum reversal, separation, and turbulence production around the inner turn.

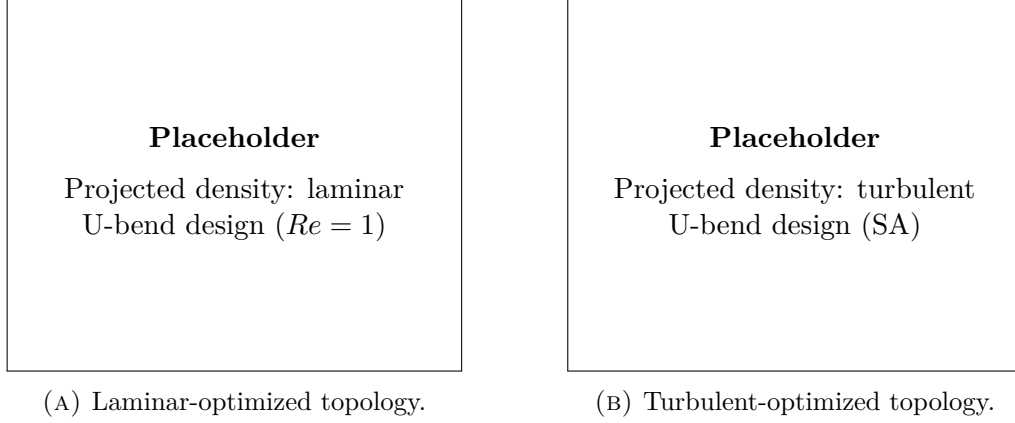


FIGURE C.1: Laminar and turbulent topology comparison for the Dilgen U-bend.

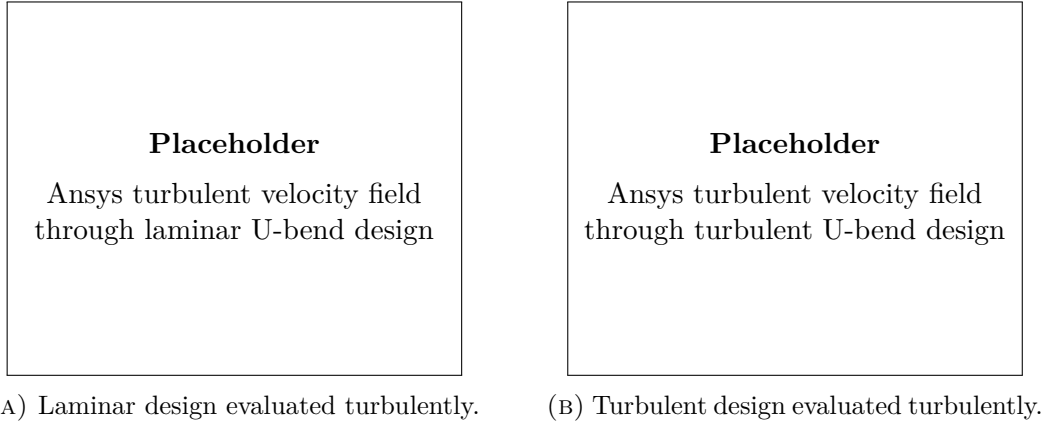


FIGURE C.2: Velocity-magnitude comparison for the extracted Dilgen U-bend designs under identical turbulent operating conditions.

TABLE C.1: Ansys mesh-independence table for the extracted laminar Dilgen U-bend design.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{\text{SST } k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{\text{SST } k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected comparison mesh		[TBD]	

TABLE C.2: Performance comparison of the extracted Dilgen U-bend topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SA	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SA	[TBD]	[TBD]	[TBD]% vs. laminar
Laminar TO, $Re = 1$	SST $k-\omega$	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SST $k-\omega$	[TBD]	[TBD]	[TBD]% vs. laminar

C.3 Alexandersen Pipe Bend

The Alexandersen pipe bend tests whether a topology optimized under turbulent conditions produces a more separation-aware turning path than a laminar topology optimized at $Re = 1$. The turbulent case in Chapter 5 uses the $V_f = 0.25$ design.

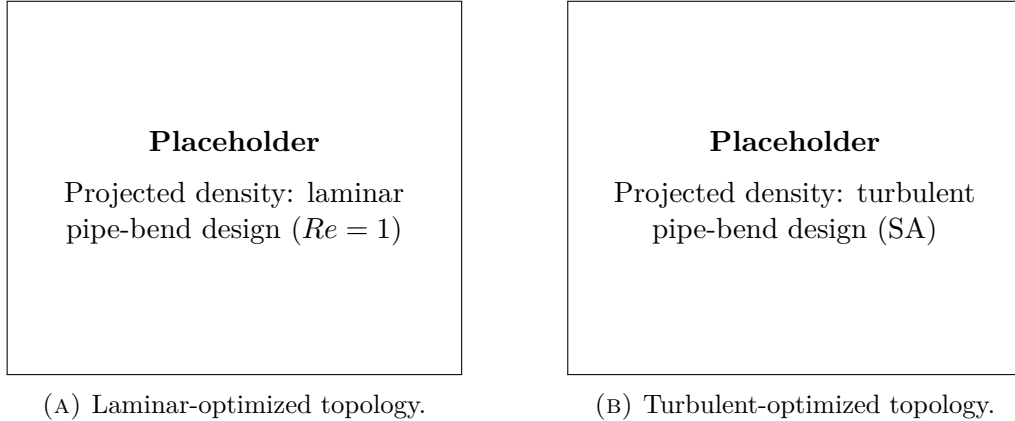
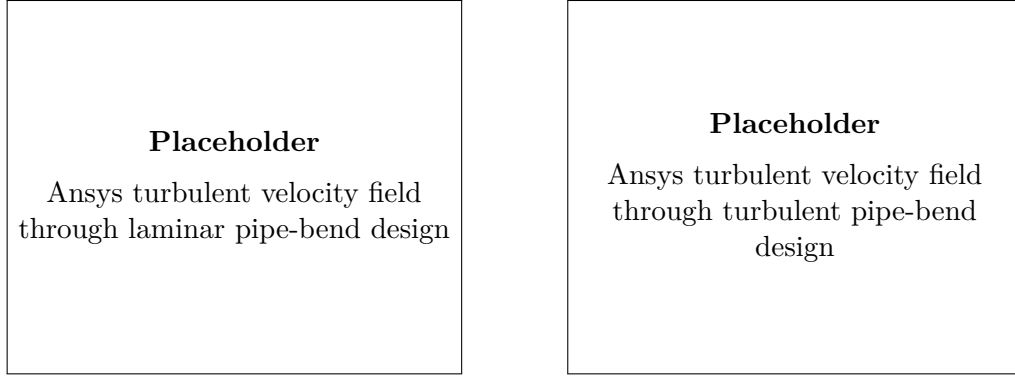


FIGURE C.3: Laminar and turbulent topology comparison for the Alexandersen pipe bend.

C. PERFORMANCE COMPARISON OF LAMINAR AND TURBULENT TOPOLOGIES



(A) Laminar design evaluated turbulently. (B) Turbulent design evaluated turbulently.

FIGURE C.4: Velocity-magnitude comparison for the extracted Alexandersen pipe-bend designs under identical turbulent operating conditions.

TABLE C.3: Ansys mesh-independence table for the extracted laminar Alexandersen pipe-bend design.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{SST\ k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{SST\ k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected comparison mesh		[TBD]	

TABLE C.4: Performance comparison of the extracted Alexandersen pipe-bend topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SA	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SA	[TBD]	[TBD]	[TBD]% vs. laminar
Laminar TO, $Re = 1$	SST $k-\omega$	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SST $k-\omega$	[TBD]	[TBD]	[TBD]% vs. laminar

C.4 Yoon Diffuser

The Yoon diffuser is expected to highlight the importance of turbulent optimization in adverse pressure gradients. A laminar diffuser topology may reduce creeping-flow dissipation but still create a separated turbulent flow when evaluated at $Re = 3,000$.

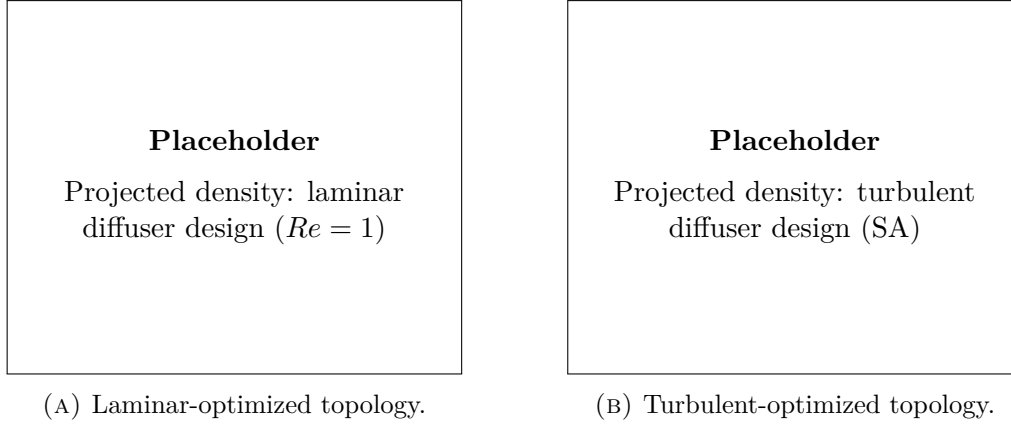


FIGURE C.5: Laminar and turbulent topology comparison for the Yoon diffuser.

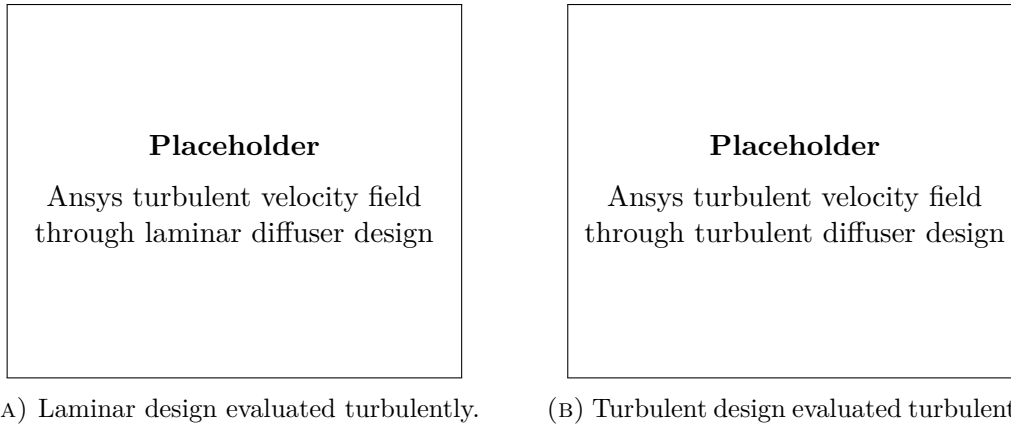


FIGURE C.6: Velocity-magnitude comparison for the extracted Yoon diffuser designs under identical turbulent operating conditions.

TABLE C.5: Ansys mesh-independence table for the extracted laminar Yoon diffuser design.

Parameter	Coarse	Medium	Fine
Bulk mesh type	[TBD]	[TBD]	[TBD]
Near-wall treatment	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
Maximum y^+	[TBD]	[TBD]	[TBD]
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\Phi_D^{SST\ k-\omega}$ [W/m]	[TBD]	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\Delta p^{SST\ k-\omega}$ [Pa]	[TBD]	[TBD]	[TBD]
Relative change, Medium–Fine		[TBD]%	
Selected comparison mesh		[TBD]	

C. PERFORMANCE COMPARISON OF LAMINAR AND TURBULENT TOPOLOGIES

TABLE C.6: Performance comparison of the extracted Yoon diffuser topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SA	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SA	[TBD]	[TBD]	[TBD]% vs. laminar
Laminar TO, $Re = 1$	SST $k-\omega$	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SST $k-\omega$	[TBD]	[TBD]	[TBD]% vs. laminar

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Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL·E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools**. More info about what sensitive and personal data are, is described in the [three confidentiality levels of the KU Leuven data classification model](#) (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**. **Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For lawfully obtained copyrighted material you should preferably use **M365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**.

- GenAI assistance may not be used for data or topics covered by a **Non-Disclosure Agreement (NDA)**. Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.
- If your master's thesis is under **embargo**, you should first discuss with your supervisor whether the use of GenAI is permitted.
- Before using a GenAI tool, always consider whether its use is responsible, including from a **sustainability perspective** (e.g. when using GenAI as a search engine, language assistant, ...).
- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.
- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.

Exam Regulations Article 84: *“Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity.”*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (**Dutch/English**).
- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (**Dutch/English**)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name:

Student number:

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

☐ This form is related to a **course assignment**.

Course name: **Course number:**

This form is related to my ☐ **Bachelor's** or ☐ **Master's thesis**.

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Daily supervisor:

Please indicate with "X":

☐ I **did not use** any GenAI assistance tool.

☐ I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot, ...):

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As a language assistant for reviewing or improving texts I wrote myself	<i>add text here</i>	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: • When using GenAI tools on texts you did not write yourself to improve the text, you have to refer to the original source or author, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	<i>add text here</i>	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. However, be careful: • If it entails text by an author other than yourself, you are not allowed to include that paraphrased text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others	<i>add text here</i>	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: • You are not allowed to include that translated text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	<i>add text here</i>	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. You only have to refer to the existing research and references you have checked and used (without such references you would be committing plagiarism and thus committing an irregularity). However, be careful: • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, verify the information and do not just copy-paste it as you should understand and critically process everything you are writing.

For literature search	<i>add text here</i>	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; you only have to refer to the literature references you have checked and used (without such references you would be committing plagiarism and thus committing an irregularity). However, be careful: • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may output no or wrong references. As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	<i>add text here</i>	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

For new ideas generating (research)	<i>add text here</i>	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. Be careful: • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. If so, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus committing an irregularity).
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For generating blocks of text (other than the allowed use without referencing mentioned above)		According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed. Be careful: • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an absolute minimum; you should always check the original sources.
For generating visuals, video or audio	<i>add text here</i>	Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. Be careful: • If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. • If you work with existing visuals, audio or video, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus an irregularity). • Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).

Other use (specify here; this may also include a combination of types of use mentioned above):	<i>add text here</i>	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis beforehand and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.
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